

ABELIAN VARIETIES AND SCHEMES

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1. PRELIMINARIES ON SCHEMES

1.1. Dedekind schemes and fibers.

Definition 1.1.1 (Dedekind scheme). A *Dedekind scheme* is a scheme S that is integral (Definition 1.2.5), Noetherian, normal, and of Krull dimension (Definition 1.2.8) 1. Equivalently, every point $s \in S$ has an open neighborhood $U = \operatorname{Spec} R$ such that R is a Dedekind domain. In particular, for every point $s \in S$, the local ring $\mathcal{O}_{S,s}$ is either a field (if s is the generic point) or a discrete valuation ring (if s is a closed point).

Proposition 1.1.2. Every DVR is a Dedekind domain. Conversely, every local Dedekind domain is a DVR.

Proof. By definition, every DVR is a PID. All PID’s are Noetherian. All PID’s are also UFD’s which are in turn integrally closed (Definition A.2.2) in their fields of fractions.

The ideals of a DVR $(R, \mathfrak{m})$ are of the form $\mathfrak{m}^n$ for some integer $n \geq 0$. The ideal $\mathfrak{m}^0$ is the unit ideal. The ideal $\mathfrak{m}^1$ is the maximal ideal. For $n \geq 2$, the ideal $\mathfrak{m}^n$ is not a prime ideal — fixing a uniformizer π of R , the element $\pi^n = \pi^{n-1} \cdot \pi$ is an element of $\mathfrak{m}^n$ and yet neither π^{n-1} nor π is an element of $\mathfrak{m}^n$.

We have thus shown that R is Noetherian, integrally closed whose only prime ideal is the maximal ideal, i.e. R is a Dedekind domain.

Conversely, suppose that R is a local Dedekind domain. Let $\mathfrak{m}$ be its maximal ideal.

(♠ TODO: verify the following converse argument) Since R is Noetherian and every nonzero prime ideal is maximal, R has Krull dimension at most 1. Because R is local, $\mathfrak{m}$ is the unique nonzero prime ideal. We first show that $\mathfrak{m}$ is principal. Let K be the field of fractions of R , and consider the fractional ideal

$$I = \{y \in K \mid y\mathfrak{m} \subseteq R\}.$$

Clearly $R \subseteq I$. Since $\mathfrak{m}$ contains a nonzero element and R is a domain, $I \neq (0)$. Moreover, for any $y \in I$, we have $y\mathfrak{m} \subseteq R$, which implies $y^2\mathfrak{m} = y(y\mathfrak{m}) \subseteq yR \subseteq I$ by induction; thus I is actually a subring of K . Because $\mathfrak{m}$ is finitely generated (as R is Noetherian), say $\mathfrak{m} = (a_1, \dots, a_n)$, and because $ya_i \in R$ for all i , we have $xI \subseteq R$ for any nonzero $x \in \mathfrak{m}$ (since $xy = yx \in y\mathfrak{m} \subseteq R$). Hence $I \subseteq x^{-1}R$. As R is Noetherian, the submodule I of the free module $x^{-1}R \cong R$ is finitely generated. For any $y \in I$, multiplication by y maps the finitely generated faithful R -module I into itself, so by the determinant trick (or Cayley-Hamilton), y satisfies a monic polynomial over R . Thus every element of I is integral over R . Since R is integrally closed in K , we conclude $I = R$. This means $\mathfrak{m}^{-1}\mathfrak{m} = R$, so $\mathfrak{m}$ is an invertible ideal.

In a local ring, every invertible ideal is principal (this follows immediately from Nakayama's Lemma: if $\mathfrak{m}/\mathfrak{m}^2$ required more than one generator over $R/\mathfrak{m}$, invertibility would fail). Hence $\mathfrak{m} = (\pi)$ for some uniformizer π .

Now let $J \neq (0)$ be any nonzero ideal of R . Since $\dim R \leq 1$, the only prime ideals are (0) and $\mathfrak{m}$, so $\sqrt{(x)} = \mathfrak{m}$ for every $0 \neq x \in J$. Thus there exists a largest integer $n \geq 0$ such that $J \subseteq \mathfrak{m}^n = (\pi^n)$. Writing $J = \pi^n K$ for some ideal K , we have $K \not\subseteq \mathfrak{m}$, which forces $K = R$ (as $\mathfrak{m}$ is the unique maximal ideal). Therefore $J = (\pi^n)$, proving that every nonzero ideal of R is principal. Consequently, R is a local PID. A local PID with Krull dimension 1 is precisely a discrete valuation ring by definition. This completes the proof. □

Definition 1.1.3 (Special fiber and generic fiber). Let S be a integral (Definition 1.2.5) scheme, let η be its generic point (Definition 1.1.4), and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let $s \in S$ be a closed point.

1. For any morphism of schemes (Definition 1.2.1) $f : X \rightarrow S$, the *generic fiber of X over S* (or simply the *generic fiber*) is the base change (Definition A.2.5):

$$X_\eta = X \times_S \operatorname{Spec} K$$

of f along $\operatorname{Spec} K \rightarrow S$.

2. For any morphism of schemes $f : X \rightarrow S$, the *special fiber of X at s* is the base change (Definition A.2.5):

$$X_s = X \times_S \operatorname{Spec} k(s)$$

of f along $\operatorname{Spec} k(s) \rightarrow S$, where $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$ is the residue field (Definition 1.2.9) at s .

In the case that S has only closed point (e.g. S is the ring of integers of a local field), the *special fiber of X* refers to the special fiber of X at this unique closed point.

Definition 1.1.4. Let X be a topological space. A point $\eta \in X$ is called a **generic point of X** if the closure of the singleton set $\{\eta\}$ is the entire space X , i.e., $\overline{\{\eta\}} = X$. More generally, if Z is an irreducible closed subset of X , a point $\eta \in Z$ is called a **generic point of Z** if $\overline{\{\eta\}} = Z$. In the context of schemes, every irreducible closed subset has a unique generic point.

Definition 1.1.5 (Closed point of a topological space). Let X be a topological space. A point $x \in X$ is called a **closed point of X** if the singleton set $\{x\}$ is closed in the topology of X , i.e., its closure satisfies $\overline{\{x\}} = \{x\}$.

In a T_1 space, every singleton set is closed, so every point is a closed point.

1.2. Basic scheme-theoretic notions.

Definition 1.2.1 (Morphism of schemes). Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be schemes. A **morphism of schemes** is a morphism as locally ringed spaces.

In particular, there is a category, often denoted by **Sch**, **Sch** etc., whose objects are schemes and whose morphisms are morphisms of schemes.

Definition 1.2.2. Let $f : X \rightarrow Y$ be a morphism of schemes (Definition 1.2.1). The morphism f is said to be **proper** if it satisfies the following three conditions:

1. f is separated,
2. f is of finite type,
3. f is universally closed.

Definition 1.2.3 (Smooth Morphism of Schemes). Let $f : X \rightarrow S$ be a morphism (Definition 1.2.1) of schemes.

We say that f is **smooth**, and that X is a **smooth scheme over S** , if it satisfies the following conditions:

- f is locally of finite presentation: for every point $x \in X$, there exists an open neighborhood $U \subseteq X$ of x and an open neighborhood $V \subseteq S$ of $f(x)$ such that the restriction $f|_U : U \rightarrow V$ corresponds to a morphism of rings $\mathcal{O}_S(V) \rightarrow \mathcal{O}_X(U)$ that is finitely presented.
- f is flat: the induced map on local rings is flat.
- For every point $x \in X$, the fiber $X_{f(x)} = X \times_S \text{Spec } \kappa(f(x))$ is a smooth variety over the residue field (Definition 1.2.9) $\kappa(f(x))$, equivalently, the sheaf of relative Kähler differentials $\Omega_{X/S}$ is locally free of finite rank.

Informally, a smooth morphism behaves like a submersion in differential geometry, providing "nice" fiber structures and descent properties.

Given a scheme S , the **category of smooth schemes over S** is the following locally small category:

- The objects are smooth morphisms $X \rightarrow S$.

- The morphisms between objects $X_1 \rightarrow S$ and $X_2 \rightarrow S$ are S -morphisms $X_1 \rightarrow X_2$ such that the following commutes:

$$\begin{array}{ccc} X_1 & \xrightarrow{\quad} & X_2 \\ & \searrow & \swarrow \\ & S & \end{array}$$

The category of smooth schemes over S is often denoted by notations such as $\mathbf{Sm}/S$, $\mathbf{Sm}/S$, $\mathbf{Sm}_S$, $\mathbf{Sm}_S$ etc.

Definition 1.2.4 (Affine scheme). Let A be a commutative ring with unity. Define the set $\mathrm{Spec}(A)$ to be the set of all prime ideals of A . Equip it with the *Zariski topology*, which is the topology whose closed sets are given by *vanishing loci*

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) : I \subseteq \mathfrak{p}\}$$

for ideals $I \subseteq A$. Define the sheaf $\mathcal{O}_{\mathrm{Spec}(A)}$, called the *structure sheaf of Spec A*, by

$$\mathcal{O}_{\mathrm{Spec}(A)}(U) = \{ \text{locally defined fractions of elements of } A \text{ on } U \},$$

for each open set $U \subseteq \mathrm{Spec}(A)$. More precisely, $\mathcal{O}_{\mathrm{Spec}(A)}(U)$ can be described as the set of functions $s : U \rightarrow \prod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$ such that

1. For each $\mathfrak{p} \in U$, the value $s(\mathfrak{p})$ lies in the localization $A_{\mathfrak{p}}$
2. For each $\mathfrak{p} \in U$, there exists an open neighborhood $V \subseteq U$ containing $\mathfrak{p}$ and elements $a, f \in A$ such that for all $\mathfrak{q} \in V$,
 - $f \notin \mathfrak{q}$, so that the fraction $\frac{a}{f}$ is well-defined in $A_{\mathfrak{q}}$.
 - $s(\mathfrak{q}) = \frac{a}{f}$ in $A_{\mathfrak{q}}$.

It is the case that the stalk at $\mathfrak{p} \in \mathrm{Spec}(A)$ is canonically the localization $A_{\mathfrak{p}}$. Then $(\mathrm{Spec}(A), \mathcal{O}_{\mathrm{Spec}(A)})$ is a locally ringed space, called the *affine scheme associated to A*.

Moreover, given $f \in A$, we define the locus $D(f)$ by

$$D(f) = \mathrm{Spec} A \setminus V((f)) = \{\mathfrak{p} \in \mathrm{Spec} A : f \notin \mathfrak{p}\}$$

Definition 1.2.5. Let X be a scheme. We say that X is an *integral scheme* if X is both a reduced scheme and an irreducible topological space. Equivalently, X is integral if for every non-empty open subset $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is an .

Definition 1.2.6 (Locally Noetherian Scheme and Noetherian Scheme). Let X be a scheme.

- X is called *locally Noetherian* if it admits an open cover $\{U_i\}$ such that for each i , the ring $\mathcal{O}_X(U_i)$ of regular functions on U_i is a Noetherian ring. Equivalently, X is locally Noetherian if it is covered by open affine subschemes $\mathrm{Spec} A_i$ with each A_i a Noetherian ring.
- X is called *Noetherian* if it is locally Noetherian and quasi-compact, i.e., X can be covered by finitely many affine opens $\mathrm{Spec} A_i$ where each A_i is Noetherian.

Definition 1.2.7. Let X be a scheme.

- We say that X is **normal at a point $x \in X$** if the local ring $\mathcal{O}_{X,x}$ is an integrally closed (Definition A.2.2) domain.
- The scheme X is called **normal** if it is normal at every point $x \in X$. If X is integral (Definition 1.2.5), this is equivalent to saying that for every affine open subset $U = \text{Spec } A$, the ring A is an integrally closed domain.
- We say that X is **factorial at a point $x \in X$** if the local ring $\mathcal{O}_{X,x}$ is a UFD.
- The scheme X is called **factorial** if it is factorial at every point.

Definition 1.2.8. Let X be a topological space. The **dimension of X** or **Krull dimension of X** , denoted by $\dim(X)$, is the supremum of the lengths of strictly ascending chains of distinct irreducible closed subsets of X . That is, $\dim(X)$ is the supremum of integers n such that there exists a chain

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$$

where each Z_i is a non-empty irreducible closed subset of X . If X is empty, its dimension is defined to be $-\infty$ (or sometimes -1).

Definition 1.2.9. Let X be a scheme and let $x \in X$ be a point. The **residue field of the scheme X at the point x** is defined as

$$\kappa(x) = \kappa(\mathcal{O}_{X,x}) := \mathcal{O}_{X,x} / \mathfrak{m}_x.$$

where $\mathcal{O}_{X,x}$ is the local ring of X at x with maximal ideal $\mathfrak{m}_x$.

When X is a scheme over a field (Definition A.1.3) k , the residue field of X at x is often denoted by $k(x)$.

Definition 1.2.10. Let R be an integral domain. Let $n \geq 1$ be an integer. The **function field over R in n variables $x_1, \dots, x_n$** is the field (Definition A.1.3) $R(x_1, \dots, x_n)$ defined as the field of fractions of the polynomial ring $R[x_1, \dots, x_n]$, which is an integral domain.

$R(x_1, \dots, x_n)$ is also called the **field of rational numbers over R in n variables**. An element of $R(x_1, \dots, x_n)$ is called a **rational function over R in n variables**.

2. ALGEBRAIC GROUP SCHEMES AND ABELIAN VARIETIES

2.1. Group schemes.

Definition 2.1.1. Let S be a scheme. An **algebraic group scheme over S** (or an **S -group scheme**) is a group object G in the category of schemes over S ; that is, G is an S -scheme equipped with S -morphisms: $m : G \times_S G \rightarrow G$ (**multiplication**), $i : G \rightarrow G$ (**inverse**), and $e : S \rightarrow G$ (**identity/identity section**), satisfying the group axioms expressed by the commutativity of the following diagrams:

$$\begin{array}{ccc}
 G \times_S G \times_S G & \xrightarrow{m \times \text{id}} & G \times_S G \\
 \text{id} \times m \downarrow & & \downarrow m \\
 G \times_S G & \xrightarrow{m} & G
 \end{array}$$

1. **Associativity**

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2. **Identity**

$$\begin{array}{ccc}
 G \times_S S & \xrightarrow{\text{id} \times e} & G \times_S G \\
 & \searrow \simeq & \downarrow m \\
 & & G
 \end{array}
 \qquad
 \begin{array}{ccc}
 S \times_S G & \xrightarrow{e \times \text{id}} & G \times_S G \\
 & \searrow \simeq & \downarrow m \\
 & & G
 \end{array}$$

3. **Inverse**

$$\begin{array}{ccc}
 G & \xrightarrow{(\text{id}, i)} & G \times_S G \\
 \text{id} \downarrow & & \downarrow m \\
 G & \xrightarrow{e \circ \pi} & G
 \end{array}$$

where $\pi : G \rightarrow S$ is the structure morphism and $e \circ \pi$ sends g to the identity section.

If the multiplication map also satisfies the commutative axiom, then the group scheme G is called *commutative* or *abelian*.

If G is affine over S , we call it an *affine group scheme over S* .

If the base scheme S is the spectrum of a field k , then we call G a *k -algebraic group* or an *algebraic group (scheme) over k* . If G is additionally a k -variety, then we call G a *k -group variety*.

Definition 2.1.2. 1. Let k be a field (Definition A.1.3), and let G be a group scheme (Definition 2.1.1) over $\text{Spec}(k)$ (Definition 1.2.4). The *identity component or neutral component of G* , denoted G^0 , is the unique connected component of G containing the identity point $e \in G(k)$.

2. Let S be a scheme, and let G be an algebraic group scheme over S . The *identity component of G* , denoted G^0 , is the unique open and closed subgroup scheme of G such that for every geometric point $\bar{s} \rightarrow S$, the fiber $(G^0)_{\bar{s}}$ is the connected component of the identity element in the group scheme $G_{\bar{s}}$.

Note that G^0/S might not be connected if S is not connected

Proposition 2.1.3 (see [Mil17, Between Definition 6.9 and Proposition 6.10]). Let G/k be a smooth algebraic group scheme over a field. The strong identity component G^{so} coincides with the identity component G^0 (Definition 2.1.2).

Definition 2.1.4. (♠ TODO: TODO: define a (commutative) group scheme) (♠ TODO: TODO: define the multiplicative group scheme) Let S be a scheme. The *additive group scheme over S* is the group scheme $\mathbb{G}_a = \mathbb{G}_{a,S}$ over S defined by

$$\mathbb{G}_{a,S} = \text{Spec } \mathcal{O}_S[T]$$

with group structure morphisms:

- **Comultiplication (Addition):** The morphism $\Delta : \mathbb{G}_{a,S} \rightarrow \mathbb{G}_{a,S} \times_S \mathbb{G}_{a,S}$ corresponding to the ring homomorphism $\mathcal{O}_S[T] \rightarrow \mathcal{O}_S[T] \otimes_{\mathcal{O}_S} \mathcal{O}_S[T]$ given by $T \mapsto T \otimes 1 + 1 \otimes T$.
- **Counit (Identity):** The morphism $\varepsilon : \mathbb{G}_{a,S} \rightarrow S$ corresponding to $T \mapsto 0$.
- **Coinverse (Inversion):** The morphism $\iota : \mathbb{G}_{a,S} \rightarrow \mathbb{G}_{a,S}$ corresponding to $T \mapsto -T$.

Thus, $(\mathbb{G}_{a,S}, \Delta, \varepsilon, \iota)$ is a commutative group scheme over S . Note that we may speak of the *additive group scheme over a ring R* as the additive group scheme over $\text{Spec } R$.

Definition 2.1.5 (Multiplicative group scheme). Let S be a scheme. The *multiplicative group scheme over S* , denoted $\mathbb{G}_{m,S}$, is the group scheme over S defined as the open subscheme

$$\mathbb{G}_{m,S} := \mathbb{A}_S^1 \setminus \{0\}_S$$

of the affine line $\mathbb{A}_S^1$, equipped with the group law given by multiplication of functions:

$$m : \mathbb{G}_{m,S} \times_S \mathbb{G}_{m,S} \rightarrow \mathbb{G}_{m,S}, \quad (x, y) \mapsto xy.$$

The identity section is the morphism

$$e : S \rightarrow \mathbb{G}_{m,S}, \quad s \mapsto 1,$$

and the inversion morphism is given by

$$i : \mathbb{G}_{m,S} \rightarrow \mathbb{G}_{m,S}, \quad x \mapsto x^{-1}.$$

Definition 2.1.6 (Kernel of a homomorphism of group schemes over a scheme). Let S be a scheme and let $\phi : G_1 \rightarrow G_2$ be a homomorphism of algebraic group schemes over S .

The *kernel of ϕ* , denoted by $\ker \phi$ or $A[\phi]$, is the fiber of ϕ at the identity section (Definition 2.1.1) $e_2 : S \rightarrow G_2$ of G_2 . Equivalently, $\ker \phi$ is the subgroup scheme of G_1 defined as the pullback:

$$\ker \phi = G_1 \times_{G_2} S,$$

(♠ TODO: finite group scheme) where the map $G_1 \rightarrow G_2$ is ϕ and the map $S \rightarrow G_2$ is the identity section. If ϕ is a finite morphism, then $\ker \phi$ is a finite group scheme over S .

Definition 2.1.7. Let S be a scheme, and let G be an algebraic group scheme over S . A *subgroup scheme* (or synonymously an *algebraic subgroup*) H of G is an S -subscheme

$$H \subseteq G$$

such that for every S -scheme T , the subset $H(T) \subseteq G(T)$ is a subgroup of the group $G(T)$. Equivalently, H is a group object in the category of S -schemes equipped with a monomorphism of group schemes $H \rightarrow G$.

Definition 2.1.8 (Quotient of an algebraic group scheme by an algebraic subgroup). Let S be a scheme. Let G be an algebraic group scheme over S (Definition 2.1.1), and let $H \subseteq G$ be a closed subgroup scheme over S , i.e., a monomorphism of group schemes $H \rightarrow G$ over S .

The *quotient sheaf* G/H is the fppf sheafification of the presheaf

$$T \mapsto G(T)/H(T)$$

on the category of S -schemes, where T is any S -scheme.

If G/H is representable by a scheme (or by an algebraic space) Q over S , then Q is called the *quotient of G by H* . In general, if H is normal, then G/H inherits the structure of a group scheme over S .

Otherwise, G/H is just a sheaf of sets (or algebraic space on $(\text{Sch}/S)_{\text{fppf}}$) over S without a natural group structure.

2.2. Algebraic tori and semi-abelian group schemes.

Definition 2.2.1 (Algebraic torus). An *(algebraic) torus over a field k* is an algebraic group (Definition 2.1.1) T over k such that the base change $T_{\bar{k}} = T \times_k \bar{k}$ is isomorphic as an algebraic group over $\bar{k}$ to $(\mathbb{G}_m/\bar{k})^n$ (Definition 2.1.5) for some nonnegative integer n . In particular, any l of k such that $T \times_k l$ is isomorphic to $(\mathbb{G}_m/l)^n$ may be called a *splitting field of the torus*. There always exists a minimal splitting field of the torus and it is a finite extension of k . Equivalently, an algebraic torus over k is an algebraic torus over $\text{Spec } k$ (Definition 1.2.4).

Definition 2.2.2 (Semi-abelian group scheme over a scheme). Let S be a scheme. A *semi-abelian (group) scheme over S* is an algebraic group scheme (Definition 2.1.1) $\pi: G \rightarrow S$ that fits into an extension of an abelian scheme by an algebraic torus, i.e., there exists a short exact sequence of algebraic groups over S :

$$1 \rightarrow T \rightarrow G \rightarrow A \rightarrow 0,$$

where $\pi_T: T \rightarrow S$ is an algebraic torus over S , $\pi_A: A \rightarrow S$ is an abelian scheme over S (Definition 4.0.1), and the maps are homomorphisms of algebraic group schemes over S . If T has rank r , we say that G has *torus rank r* . When $r = 0$, the semi-abelian group scheme is simply an abelian scheme.

2.3. Abelian varieties.

Definition 2.3.1 (Abelian variety over a field). Let k be a field. An *abelian variety over k* is a complete, connected algebraic group variety defined over k , i.e.

- A is a smooth (Definition 1.2.3), proper (Definition 1.2.2), geometrically connected (Definition A.2.6) algebraic variety over k ,
- A is endowed with a group structure defined by morphisms of varieties over k (multiplication $m: A \times_k A \rightarrow A$ and inverse $i: A \rightarrow A$),
- the group law satisfies the group axioms scheme-theoretically.

In particular, an abelian variety is a projective algebraic group variety (Definition 2.1.1) over k .

Definition 2.3.2 (Dual abelian variety). Let A be an abelian variety (Definition 2.3.1) over a field k . The *dual abelian variety of A* , denoted by A^* , is the abelian variety representing the Picard functor $\text{Pic}_{A/k}$, i.e., (♠ TODO: picard functor)

$$A^* = \text{Pic}^0(A),$$

where $\text{Pic}^0(A)$ is the connected component of the identity in the Picard scheme of A . For any commutative k -algebra R , the group $A^*(R)$ classifies line bundles on $A \times_k \text{Spec } R$ that have degree zero on every fiber over $\text{Spec } R$.

Definition 2.3.3. Let A and B be abelian varieties over a field k . We say that A and B are *isogenous*, often written $A \sim B$, if there exists an k -isogeny (Definition 4.0.2) $A \rightarrow B$. The relation $\sim$ turns out to be an equivalence relation on the set of abelian varieties over a fixed field k .

Definition 2.3.4 (Isogeny of abelian varieties over a field). (♠ TODO: finite group scheme) Let k be a field (Definition A.1.3) and let A and B be abelian varieties (Definition 2.3.1) over k . An *isogeny of abelian varieties over k* is a homomorphism $\varphi: A \rightarrow B$ of group schemes over k that is surjective and has finite kernel (Definition 2.1.6). The kernel $\ker \varphi$ is a finite group scheme over k (Definition 2.1.6). Equivalently, φ is a finite, faithfully flat homomorphism of abelian varieties over k .

Definition 2.3.5 (Polarization of an abelian variety). Let A be an abelian variety (Definition 2.3.1) over a field (Definition A.1.3) k . Let $A^* = \text{Pic}^0(A)$ be the dual abelian variety (Definition 2.3.2) of A . For any line bundle L on A , define $\phi_L: A \rightarrow A^*$ by

$$\phi_L(x) = t_x^* L \otimes L^{-1},$$

where $t_x: A \rightarrow A$ is translation by x . The map ϕ_L is a homomorphism of abelian varieties. A *polarization on A* is an isogeny $\phi: A \rightarrow A^*$ such that there exists an ample line bundle L on A with $\phi = \phi_L$. When this holds, we say that (A, ϕ) is a *polarized abelian variety*.

Definition 2.3.6 (Principal polarization of an abelian variety). Let (A, ϕ) be a polarized abelian variety (Definition 2.3.5) over a field (Definition A.1.3). The polarization ϕ is called *principal* if it is an isomorphism (equivalently, if the degree of ϕ , denoted by $\deg(\phi)$, equals 1). An abelian variety equipped with a principal polarization is called a *principally polarized abelian variety* (PPAV).

For $g = 1$, a PPAV is an elliptic curve. A principally polarized abelian surface ($g = 2$) or threefold ($g = 3$) is the Jacobian of a smooth connected algebraic curve unless it decomposes as a product together with the product polarization.

Definition 2.3.7 (Weil–Châtelet group of an abelian variety over a field). Let k be a field, and let A be an abelian variety defined over k . Denote by $G_k = \text{Gal}(\bar{k}/k)$ the absolute Galois group of k .

The *Weil–Châtelet group of A over k* , denoted $\text{WC}(A/k)$, is defined as the collection equivalence classes of homogeneous spaces for A/k .

2.4. Endomorphism rings and algebras.

Definition 2.4.1 (Endomorphism algebra up to isogeny of an abelian variety over a field). Let k be a field and let A be an abelian variety over k (Definition 2.3.1). The *endomorphism ring of A* , denoted by $\text{End}(A)$, is the set of morphisms of abelian varieties over k from A to itself. With addition given by the group law on A and multiplication given by composition, $\text{End}(A)$ forms a ring. The *endomorphism algebra up to isogeny of A* , denoted by $\text{End}^0(A)$, is defined by

$$\text{End}^0(A) = \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Definition 2.4.2 (Simple abelian variety). Let k be a field (Definition A.1.3) and let A be an abelian variety over k (Definition 2.3.1). We say that A is a *simple abelian variety over k* if $\dim A > 0$ and the only closed subgroup schemes of A which are themselves abelian varieties over k (Definition 2.3.1) are the trivial subgroup scheme $\{0_A\}$ and A itself. Equivalently, every abelian subvariety of A is either 0 or A .

Definition 2.4.3. Let k be a field (Definition A.1.3), let $\bar{k}$ be an algebraic closure of k , and let A be an abelian variety over k (Definition 2.3.1). Let $A_{\bar{k}} = A \times_k \bar{k}$ denote the base change (Definition A.2.5) of A to $\bar{k}$.

We say that A is an *absolutely simple abelian variety over k* (or a *geometrically simple abelian variety over k*) if $A_{\bar{k}}$ is a simple abelian variety over $\bar{k}$ (Definition 2.4.2).

Equivalently, A is absolutely simple if $\dim A > 0$ and the only closed subgroup schemes of $A_{\bar{k}}$ which are abelian varieties over $\bar{k}$ are the trivial subgroup scheme $\{0_{A_{\bar{k}}}\}$ and $A_{\bar{k}}$ itself.

Theorem 2.4.4 (Albert's classification of simple abelian varieties). (♠ TODO: Rosati involution, End0,) Let A be a simple abelian variety (Definition 2.4.2) over a field (Definition A.1.3) k . Then $\text{End}^0(A) = \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ is a central division algebra over $\mathbb{Q}$, and any Rosati involution (associated to a principal polarization on A) is a positive involution. Let $D = \text{End}^0(A)$, let $*$ denote the Rosati involution, and set $F = D^*$, the fixed field of $*$ in D . Then exactly one of the following four cases holds:

1. **Type I (Totally real):** $D = F$ is a totally real number field with $[F : \mathbb{Q}] = \dim A$, and $*$ is the identity on F .
♠ TODO: totally definite quaternion algebra
2. **Type II (Totally definite quaternion over a totally real field):** D is a totally definite quaternion algebra over its center F , a totally real number field of degree $[F : \mathbb{Q}] = d$, with $*$ inducing the identity on F . We have $\dim A$ divisible by $2d$.
3. **Type III (Totally indefinite quaternion over a totally real field):** D is a totally indefinite quaternion algebra over its center F , a totally real number field, with $*$ inducing the identity on F . Here $\dim A$ is even.
4. **Type IV (Central division algebra over a CM field):** D is a central division algebra over a CM field (Definition 2.5.1) E of degree $[E : \mathbb{Q}] = 2d$, with $*$ inducing the complex conjugation on E . Let e be the index of D (i.e. $\dim_E D = e^2$). Then there exists a positive integer m such that $\dim A = m \cdot [E : \mathbb{Q}] \cdot e/2$.

Conversely, for each type and each admissible combination of invariants, there exists a simple abelian variety over some field whose endomorphism algebra realizes the given structure.

Definition 2.4.5 (Abelian variety with real multiplication over a field). Let k be a field (Definition A.1.3) and let A be an abelian variety over k (Definition 2.3.1) of dimension g . We say that A has *real multiplication (RM)* if there exists a commutative semisimple $\mathbb{Q}$ -subalgebra $F \subset \text{End}^0(A)$ (Definition 2.4.1) such that $[F : \mathbb{Q}] = g$, where $\text{End}^0(A) = \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ is the endomorphism algebra up to isogeny (Definition 2.4.1).

If A is simple (Definition 2.4.2), then F is a *totally real number field* (totally real and totally imaginary number fields) and it is the full endomorphism algebra, i.e. $\text{End}^0(A) = F$. In general, F decomposes as a product of totally real number fields corresponding to the simple factors of A up to isogeny (Definition 2.3.4).

Definition 2.4.6 (Abelian variety with quaternionic multiplication over a field). Let k be a field (Definition A.1.3) and let A be an abelian variety over k (Definition 2.3.1) of dimension g . We say that A has *quaternionic multiplication (QM)* if there exists a quaternion algebra

D over $\mathbb{Q}$ with an embedding of $\mathbb{Q}$ -algebras $D \hookrightarrow \text{End}^0(A)$, where $\text{End}^0(A) = \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ is the endomorphism algebra up to isogeny (Definition 2.4.1). In this case, g must be even.

If A is simple (Definition 2.4.2), then D is a totally definite quaternion algebra over $\mathbb{Q}$, i.e. D is a division algebra, and it is the full endomorphism algebra, i.e. $\text{End}^0(A) = D$. In particular, $[D : \mathbb{Q}] = 4$ and $g = 2$.

2.5. Complex multiplication theory.

Definition 2.5.1 (CM Field). Let K be a number field. The field K is called a **CM field** if it is a totally imaginary quadratic extension of a totally real subfield F . That is, there exists a totally real number field $F \subset K$ such that $[K : F] = 2$ and K is totally imaginary.

Every CM field has a canonical involution called the **complex conjugation**, which is the unique nontrivial automorphism of K/F when F is the maximal totally real subfield of K .

Definition 2.5.2 (Complex multiplication of an abelian variety over a field). Let k be a field (Definition A.1.3) and let A be an abelian variety over k (Definition 2.3.1) of dimension g . We say that A has **complex multiplication (CM)** if there exists a commutative semisimple $\mathbb{Q}$ -subalgebra $E \subset \text{End}^0(A)$ (Definition 2.4.1) such that $[E : \mathbb{Q}] = 2g$, where $\text{End}^0(A) = \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ is the endomorphism algebra up to isogeny (Definition 2.4.1).

If E is a CM field (Definition 2.5.1), then we say that A has **CM by E** . Whenever A is simple (Definition 2.4.2), E is a CM field (Definition 2.5.1) and it is the full endomorphism algebra, i.e. $\text{End}^0(A) = E$. In general, E decomposes as a product of CM fields corresponding to the simple factors of A up to isogeny (Definition 2.3.4).

Definition 2.5.3 (CM Type and Reflex Field of a CM Type). (**♠ TODO: AI generated, verify**) Let E be a CM field (Definition 2.5.1) with maximal totally real subfield F , and let $\text{Hom}(E, \overline{\mathbb{Q}})$ denote the set of embeddings of E into a fixed algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$. Complex conjugation on $\mathbb{C}$ induces an involution on $\text{Hom}(E, \overline{\mathbb{Q}})$: for $\tau \in \text{Hom}(E, \overline{\mathbb{Q}})$, write $\bar{\tau} := c \circ \tau$, where c is the complex conjugation (Definition 2.5.1) automorphism of E .

A **CM type for E** is a subset $\Phi \subset \text{Hom}(E, \overline{\mathbb{Q}})$ such that $|\Phi| = [E : F]$ and Φ contains exactly one element from each conjugate pair $\{\tau, \bar{\tau}\}$ in $\text{Hom}(E, \overline{\mathbb{Q}})$. Equivalently,

$$\text{Hom}(E, \overline{\mathbb{Q}}) = \Phi \sqcup \bar{\Phi}.$$

Let (E, Φ) be a CM type. The **reflex field of the CM type (E, Φ)** , denoted by $E^*(\Phi)$, is the subfield of $\overline{\mathbb{Q}}$ generated over $\mathbb{Q}$ by the traces

$$\left\{ \sum_{\tau \in \Phi} \tau(\alpha) \mid \alpha \in E \right\}.$$

Definition 2.5.4. (**♠ TODO: AI generated, verify**) Let E be a CM field (Definition 2.5.1) with maximal totally real subfield F . Let $\mathbb{G}_m = \mathbb{G}_{m, \mathbb{Q}}$ be the multiplicative group scheme (Definition 2.1.5) over $\mathbb{Q}$, and let $\text{Res}_{L/K}$ denote the Weil restriction of scalars from a number field K to another whenever $K \subseteq L$. (**♠ TODO: weil restriction**) For any number field L

containing E , write $X(L) = \text{Hom}(L, \overline{\mathbb{Q}})$ for the set of embeddings of L into $\overline{\mathbb{Q}}$, and let $T_L := \text{Res}_{L/\mathbb{Q}}(\mathbb{G}_m)$ be the associated algebraic torus over $\mathbb{Q}$. The character group of T_L is canonically isomorphic to the permutation module $\mathbb{Z}[X(L)]$.

Let (E, Φ) be a CM type (Definition 2.5.3) for E , and let E^* be the associated reflex field (Definition 2.5.3). For each embedding $\rho \in X(E^*)$, define

$$n_\rho := |\{\tau \in \Phi : \tau|_{E^*} = \rho\}|$$

to be the number of embeddings in Φ whose restriction to E^* equals ρ . Note that $n_\rho \geq 0$ and $\sum_{\rho \in X(E^*)} n_\rho = |\Phi| = [E : F]$.

The *reflex norm associated to (E, Φ)* , denoted $N_{(E, \Phi)} : T_{E^*} \rightarrow T_E$, is the unique morphism of algebraic tori (Definition 2.2.1) over $\mathbb{Q}$ whose dual map on character groups (♠ TODO: dual map on characters groups of map of algebraic tori)

$$N_{(E, \Phi)}^* : \mathbb{Z}[X(E)] \rightarrow \mathbb{Z}[X(E^*)]$$

sends each basis element $\tau \in X(E)$ to

$$N_{(E, \Phi)}^*(\tau) = n_{\tau|_{E^*}} \cdot (\tau|_{E^*}),$$

where $n_{\tau|_{E^*}} = |\{\sigma \in \Phi : \sigma|_{E^*} = \tau|_{E^*}\}|$ is the number of embeddings in Φ that have the same restriction to E^* as τ .

Equivalently, for each $\rho \in X(E^*)$, one has

$$N_{(E, \Phi)}^* \left(\sum_{\tau \in X(E), \tau|_{E^*} = \rho} \tau \right) = n_\rho \cdot \rho.$$

The kernel of the reflex norm $N_{(E, \Phi)} : T_{E^*} \rightarrow T_E$ is an algebraic subgroup of T_{E^*} whose group of connected components is a finite abelian group that appears in bounds on Galois images associated to CM abelian varieties with CM type (E, Φ) .

Relation to the field norm. When (E, Φ) is the unique CM type for a quadratic imaginary extension (i.e., $[E : F] = 1$), we have $E^* = F$ and $n_\rho = 1$ for all ρ , so the reflex norm coincides with the morphism of tori induced by the field norm $N_{E/F} : E^\times \rightarrow F^\times$.

Proposition 2.5.5 (Reflex Field Degree Formula for a CM Type). Let E be a CM field (Definition 2.5.1) with maximal totally real subfield F , and let (E, Φ) be a CM type (Definition 2.5.3). Let $g = [E : F]$ and let $E^*(\Phi)$ be the reflex field (Definition 2.5.3) of (E, Φ) .

Let $\Gamma = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ act on $\text{Hom}(E, \overline{\mathbb{Q}})$ by right multiplication: $(\tau^\sigma)(x) = \sigma(\tau(\sigma^{-1}(x)))$ for $\sigma \in \Gamma$, $\tau \in \text{Hom}(E, \overline{\mathbb{Q}})$, $x \in E$. Let $\Gamma_\Phi \subset \Gamma$ be the stabilizer of Φ . Then

$$[E^*(\Phi) : \mathbb{Q}] = [\Gamma : \Gamma_\Phi].$$

Let $h(\Phi) := |\{\tau \in \Phi : \tau|_{E^*(\Phi)} = \rho\}|$ for any (equivalently, every) $\rho \in \text{Hom}(E^*(\Phi), \overline{\mathbb{Q}})$. Then $h(\Phi)$ divides $2g$, and

$$[E^*(\Phi) : \mathbb{Q}] \cdot h(\Phi) = 2g.$$

Proposition 2.5.6 (Reflex of the Reflex: Shimura–Tate Duality for CM Types). Let (E, Φ) be a CM type (Definition 2.5.3), and let $E^*(\Phi)$ be the reflex field (Definition 2.5.3).

There exists a CM type Φ^* for $E^*(\Phi)$ such that $(E^*(\Phi))^*(\Phi^*)$ is canonically identified with E over $\mathbb{Q}$, and the reflex of $(E^*(\Phi), \Phi^*)$ recovers (E, Φ) . In particular, the reflex operation is an involution on the set of isomorphism classes of CM types.

Proposition 2.5.7 (Reflex Field Characterization via Tori and the Reflex Norm). Let (E, Φ) be a CM type (Definition 2.5.3), and let $E^*(\Phi)$ be the reflex field (Definition 2.5.3). Let $\mathbb{G}_m = \mathbb{G}_{m, \mathbb{Q}}$ and for any number field L , let $T_L = \text{Res}_{L/\mathbb{Q}}(\mathbb{G}_m)$ be the associated algebraic torus over $\mathbb{Q}$ whose character group is the permutation module $X(T_L) \cong \text{Hom}(L, \overline{\mathbb{Q}})$.

The reflex field $E^*(\Phi)$ is the unique number field (up to $\mathbb{Q}$ -isomorphism) for which there exists a morphism of algebraic tori over $\mathbb{Q}$, called the reflex norm (Definition 2.5.4),

$$N_{(E, \Phi)} : T_{E^*(\Phi)} \rightarrow T_E,$$

whose dual map on character groups

$$N_{(E, \Phi)}^* : X(T_E) \rightarrow X(T_{E^*(\Phi)})$$

sends each $\tau \in \text{Hom}(E, \overline{\mathbb{Q}})$ to

$$N_{(E, \Phi)}^*(\tau) = h_\rho \cdot \rho,$$

where $\rho = \tau|_{E^*(\Phi)} \in \text{Hom}(E^*(\Phi), \overline{\mathbb{Q}})$ and $h_\rho = |\{\sigma \in \Phi : \sigma|_{E^*(\Phi)} = \rho\}|$.

Proposition 2.5.8 (Reflex Field of an Imaginary Quadratic CM Type is $\mathbb{Q}$). Let E be an imaginary quadratic field with unique CM type (Definition 2.5.3) $\Phi = \{\tau\}$ (up to complex conjugation). Then the reflex field (Definition 2.5.3) is $E^*(\Phi) = \mathbb{Q}$.

The reflex norm (Definition 2.5.4) $N_{(E, \Phi)} : T_{\mathbb{Q}} \rightarrow T_E$ coincides with the morphism of tori induced by the field norm $N_{E/\mathbb{Q}} : E^\times \rightarrow \mathbb{Q}^\times$.

2.6. Structural properties of CM abelian varieties.

Proposition 2.6.1. Let A/K be a CM abelian variety (Definition 2.5.2) of dimension g over a field (Definition A.1.3) with complex multiplication by a CM field (Definition 2.5.1) E and assume that E is of dimension $2g$, i.e. $\text{End}_K(A) \otimes \mathbb{Q} = E$ (Definition 2.4.1). Let ℓ be a prime number.

1. The rational ℓ -adic Tate module (Definition 7.2.1) $V_\ell(A)$ is a free $E \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ -module of rank 1.
2. The integral ℓ -adic Tate module (Definition 7.2.1) $T_\ell(A)$ is a locally free $\text{End}_K(A) \otimes \mathbb{Z}_\ell$ -module of rank 1.

In particular, if $\text{End}_K(A) \cong \mathcal{O}_E$ where $\mathcal{O}_E$ is the ring of integers of E , then $T_\ell(A)$ is a free $\mathcal{O}_E \otimes \mathbb{Z}_\ell$ -module of rank 1.

Definition 2.6.2. Let A/K be a CM abelian variety (Definition 2.5.2) of dimension g over a field (Definition A.1.3) with complex multiplication by a CM field (Definition 2.5.1) E and assume that E is of dimension $2g$, i.e. $\text{End}_K(A) \otimes \mathbb{Q} = E$ (Definition 2.4.1). Let ℓ be a prime number.

1. Recalling that (Proposition 2.6.1) the integral ℓ -adic Tate module (Definition 7.2.1) $T_\ell(A)$ is a locally free $\text{End}_K(A) \otimes \mathbb{Z}_\ell$ -module of rank 1, one can identify $(\text{End}_K(A) \otimes \mathbb{Z}_\ell)^\times$ as a subgroup of $\text{Aut}(T_\ell(A))$ and the image of the integral ℓ -adic Galois representation

$$\rho_{A,\ell^\infty} : \text{Gal}(\overline{K}/K) \rightarrow \text{Aut}(T_\ell(A))$$

is contained in $(\text{End}_K(A) \otimes \mathbb{Z}_\ell)^\times$.

2. The image of the rational ℓ -adic Galois representation

$$\rho_{A,\ell^\infty} : \text{Gal}(\overline{K}/K) \rightarrow \text{Aut}(V_\ell(A))$$

is contained in $E^\times$.

2.7. CM abelian varieties and Galois representations.

Theorem 2.7.1 ([Lom17, Proposition 1.6]). Let A/K be an absolutely simple abelian variety (Definition 2.4.3) over a number field admitting over K complex multiplication (Definition 2.5.2) by a CM field (Definition 2.5.1) E , and let ℓ be a prime number that is unramified in E . (♠ TODO: unramified) (♠ TODO: reflex field) Let E^* be the reflex field of E and suppose that A has good reduction at all places of K of characteristic ℓ . Write G_ℓ for the image of the mod- ℓ Galois representation

$$\text{Gal}(\overline{K}/K) \rightarrow \text{Aut}(A[\ell]).$$

(Definition 7.1.2) (♠ TODO: how does the intersection make sense? Do the two groups live in some ambient space?) (♠ TODO: TE, reflex norm) Write F for the group of connected components of the kernel of the reflex norm $T_{E^*} \rightarrow T_E$. (♠ TODO: should discuss notion of $\text{MT}(A)(\mathbb{F}_\ell)$, which is non-trivial definition of Ono.)

1. The index $[\text{MT}(A)(\mathbb{F}_\ell) : G_\ell \cap \text{MT}(A)(\mathbb{F}_\ell)]$?? divides $[K : E^*] \cdot |F|$.
2. If ℓ is unramified in K as well, then $[\text{MT}(A)(\mathbb{F}_\ell) : G_\ell \cap \text{MT}(A)(\mathbb{F}_\ell)]$ divides $|F|$.

Lemma 2.7.2 (See [Lom17, 6.4.]). Let A/K be an absolutely simple abelian variety (Definition 2.4.3) over a number field with complex multiplication (Definition 2.5.2) by a CM field (Definition 2.5.1) E .

The kernel of the reflex norm (Definition 2.5.4) associated to a CM type (Definition 2.5.3) (E, Φ) with reflex field (Definition 2.5.3) E^* is connected.

2.8. Hodge-theoretic perspective on reflex fields.

Definition 2.8.1 (Rational Hodge Type of a Complex Abelian Variety). (♠ TODO: AI generated, verify) Let A be an abelian variety (Definition 2.3.1) over $\mathbb{C}$ of dimension g , and let $D = \text{End}^0(A)$ be its rational endomorphism algebra (Definition 2.4.1). The first rational homology $V := H_1(A(\mathbb{C}), \mathbb{Q})$ carries a pure weight-1 Hodge structure $(V, V^{1,0} \oplus V^{0,1})$, with $V_{\mathbb{R}} = V^{1,0} \oplus V^{0,1}$ and $\dim_{\mathbb{C}} V^{1,0} = \dim_{\mathbb{C}} V^{0,1} = g$.

The action of D on V extends to $D_{\mathbb{R}} := D \otimes_{\mathbb{Q}} \mathbb{R}$, and the Hodge decomposition determines a cocharacter

$$\xi : \mathbb{G}_{m,\mathbb{R}} \rightarrow \text{GL}(V_{\mathbb{R}}) \subset D_{\mathbb{R}}$$

such that $z \in \mathbb{C}^\times = \mathbb{G}_m(\mathbb{C})$ acts on $V^{p,q}$ via the character $z^{-p}\bar{z}^{-q}$. The $\mathrm{GL}(V)$ -conjugacy class of ξ is defined over $\mathbb{Q}$; its image in $D_{\mathbb{R}}$ determines a $D(\mathbb{Q})$ -conjugacy class of cocharacters

$$\xi : \mathbb{G}_{m,\mathbb{R}} \rightarrow D_{\mathbb{R}}.$$

This conjugacy class is called the *rational Hodge type of A* .

Equivalently, for each real embedding $\sigma : D \hookrightarrow D_{\mathbb{R}}$, the rational Hodge type specifies a subset $\Phi_\sigma \subset \mathrm{Hom}_{\mathbb{R}}(D_{\mathbb{R}}, \mathbb{R})$ describing which linear functionals on $D_{\mathbb{R}}$ appear as weight- $(1, 0)$ components. The collection of these subsets across all real embeddings encodes the same data as ξ .

Definition 2.8.2 (Reflex Field of the Rational Hodge Type of a Complex Abelian Variety). (♠ TODO: AI generated, verify) Let A be an abelian variety (Definition 2.3.1) over $\mathbb{C}$ of dimension g , with rational endomorphism algebra (Definition 2.4.1) $D = \mathrm{End}^0(A)$. Let $\xi : \mathbb{G}_{m,\mathbb{R}} \rightarrow D_{\mathbb{R}}$ be the rational Hodge type (Definition 2.8.1) of A , a $D(\mathbb{Q})$ -conjugacy class of cocharacters.

The *reflex field of the rational Hodge type ξ* , denoted by $E^*(\xi)$, is the smallest number field over which the conjugacy class of ξ becomes rational when viewed as a cocharacter of the algebraic group $D_{\mathbb{R}}$. Equivalently, it is the field of moduli of the Hodge structure on $H_1(A(\mathbb{C}), \mathbb{Q})$ equipped with the action of D .

When $D = E$ is a CM field (Definition 2.5.1) and ξ corresponds to a CM type (Definition 2.5.3) (E, Φ) , the reflex field $E^*(\xi)$ coincides with the reflex field of the CM type (E, Φ) (Definition 2.5.3).

Remark 2.8.3 (Generalization of Reflex Fields Beyond CM Fields). The notion of reflex field of a CM type (Definition 2.5.3) generalizes to abelian varieties with other endomorphism types via the rational Hodge type (Definition 2.8.1). See Definition 2.8.2.

This construction applies when $\mathrm{End}^0(A)$ is any semisimple $\mathbb{Q}$ -algebra satisfying $[\mathrm{End}^0(A) : \mathbb{Q}] \leq 2g$, including:

- $\mathrm{End}^0(A) = E$ a CM field (Definition 2.5.1) (the classical case);
- $\mathrm{End}^0(A) = F$ a totally real field with $[F : \mathbb{Q}] = g$ (real multiplication (Definition 2.4.5));
- $\mathrm{End}^0(A)$ a quaternion algebra over a totally real field;
- $\mathrm{End}^0(A)$ an Albert type IV division algebra.

In the RM case, the Hodge type is unique up to conjugacy (there is only one choice of $\{1, 0\}$ -grading on each real embedding), and the reflex field equals the maximal totally real subfield.

3. PICARD GROUPS AND JACOBIAN VARIETIES

Definition 3.0.1. (♠ TODO: picard group of an algebraic space) Let X be a scheme. The *Picard group of X* , denoted by $\mathrm{Pic}(X)$, is the group of isomorphism classes of invertible

sheaves (Definition A.3.5) (line bundles) on X (with respect to the Zariski topology), with the group operation induced by the tensor product of sheaves (Definition A.3.6).

More precisely:

1. Two invertible sheaves (Definition A.3.1) $\mathcal{L}$ and $\mathcal{M}$ are isomorphic if there exists an $\mathcal{O}_X$ -module (Definition A.3.4) isomorphism $\phi : \mathcal{L} \xrightarrow{\sim} \mathcal{M}$.
2. The tensor product (Definition A.3.6) of invertible sheaves induces a well-defined group operation on the set of isomorphism classes. (♠ TODO: sheafy dual)
3. The inverse of an invertible sheaf $\mathcal{L}$ is given by its dual $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$ (Definition A.3.7).
4. The identity element is the class of the structure sheaf $\mathcal{O}_X$.

(♠ TODO: class group of ring) When $X = \text{Spec}(A)$ is an affine scheme (Definition 1.2.4), there is a canonical isomorphism

$$\text{Pic}(\text{Spec}(A)) \cong \text{Cl}(A),$$

where $\text{Cl}(A)$ is the class group of the ring A (the group of isomorphism classes of rank-1 projective modules over A , also known as the Picard group of the ring).

(♠ TODO: extract this out) If X is a Noetherian, integral (Definition 1.2.5), and locally factorial scheme, then $\text{Pic}(X)$ is isomorphic to the Weil divisor class group $\text{Cl}(X)$.

3.1. The relative Picard functor of a scheme over a scheme.

Definition 3.1.1. Let X be a scheme over a scheme S . The fppf sheaf (Definition A.3.1) associated to the functor

$$P_{X/S} : (\text{Sch}/S)^\circ \rightarrow \mathbf{Sets}, \quad T \mapsto \text{Pic}(X \times_S T)$$

is denoted by $\text{Pic}_{X/S}$ and is called the *relative Picard functor of X over S* .

For an S -scheme T , the group $\text{Pic}_{X/S}(T)$ is called the *relative Picard group of $X \times_S T$ over T* .

Proposition 3.1.2 (see [BLR12, Chapter 8, after Definition 2 and after Proposition 3]). Let $f : X \rightarrow S$ be a scheme over a scheme. The relative Picard functor (Definition 3.1.1)

$$\text{Pic}_{X/S} : (\text{Sch}/S)_{\text{fppf}}^\circ \rightarrow \mathbf{Sets}$$

can be constructed as

$$\text{Pic}_{X/S}(T) = H^0(T, R^1 f_*(\mathbb{G}_m))$$

(Definition A.3.8).

If f is proper, then this formula remains valid when H^0 and $R^1 f_*(\mathbb{G}_m)$ are computed via the étale topology.

Proposition 3.1.3. Let X be a scheme. The sheaf cohomology groups

$$H^1(X, \mathbb{G}_m)$$

(Definition A.3.8) with respect to the (big) fpqc, fppf, étale, and Zariski topologies are all naturally isomorphic to one another and are isomorphic to the Picard group of X .

3.2. Representability of the relative Picard functor.

Theorem 3.2.1 ([Gro66, n°232, Théorème 3.1], see [BLR12, Section 8.2, Theorem 1]). (♠ **TODO: projective morphism of schemes**) Let $f : X \rightarrow S$ be a projective, finitely presented, flat morphism of schemes (Definition 1.2.1) whose geometric fibers (Definition A.2.4) are integral (Definition 1.2.5). The fppf-sheaf (Definition A.3.1)

$$\mathrm{Pic}_{X/S} : (\mathrm{Sch}/S)_{\mathrm{fppf}}^{\circ} \rightarrow \mathbf{Sets}$$

(Definition 3.1.1) is representable by a separated S -scheme which is locally of finite presentation over S .

Theorem 3.2.2 (Mumford, unpublished, see [BLR12, Section 8.2, Theorem 2]). Let $f : X \rightarrow S$ be flat, projective, and finitely presented with geometrically reduced fibres. Assume that the irreducible components of the fibres of f are geometrically irreducible.

The fppf-sheaf (Definition A.3.1)

$$\mathrm{Pic}_{X/S} : (\mathrm{Sch}/S)_{\mathrm{fppf}}^{\circ} \rightarrow \mathbf{Sets}$$

(Definition 3.1.1) is representable by a (not necessarily separated) S -scheme which is locally of finite presentation over S .

Theorem 3.2.3 ([Mur64], [Oor62], see [BLR12, Section 8.2, Theorem 3]). Let X be a proper (Definition 1.2.2) scheme over a field (Definition A.1.3) k . The fppf-sheaf (Definition A.3.1)

$$\mathrm{Pic}_{X/k} : (\mathrm{Sch}/k)_{\mathrm{fppf}}^{\circ} \rightarrow \mathbf{Sets}$$

(Definition 3.1.1) is representable by a locally of finite type scheme over k .

Proposition 3.2.4 ([BLR12, Section 8.4, Proposition 2]). Let $f : X \rightarrow S$ be a proper (Definition 1.2.2), flat, locally of finite presentation morphism of schemes (Definition 1.2.1). Let s be a point of S such that $H^2(X_s, \mathcal{O}_{X_s}) = 0$. (♠ **TODO: for what topology**) Then there exists an open neighborhood U of s such that $\mathrm{Pic}_{X/S}|_U$ (Definition 3.1.1) is formally smooth over U . (♠ **TODO: formally smooth**) (♠ **TODO: a theorem that says that sheaf cohomology for coherent sheaves on the zariski, etale, fppf topologies are all the same.**)

Definition 3.2.5. Let S be an algebraic stack.

3.3. Picard variety.

Definition 3.3.1. Let X/k be a smooth, geometrically integral projective variety over a field (Definition A.1.3) k .

The relative Picard functor (Definition 3.1.1) $\mathrm{Pic}_{X/k}$ is representable by a smooth scheme (Definition 1.2.3) over k . (♠ **TODO: show that the identity component is an abelian variety**) The identity component (Definition 2.1.2) $\mathrm{Pic}_{X/k}^0$ is in fact an abelian variety and is called the *Picard variety of X/k* .

(♠ **TODO: representability by algebraic space**)

3.4. Jacobian variety of a curve.

Theorem 3.4.1. Let X be a proper (Definition 1.2.2) curve over a field (Definition A.1.3) k . The relative Picard functor (Definition 3.1.1) $\mathrm{Pic}_{X/k}$ is representable by a smooth scheme (Definition 1.2.3) over k .

Definition 3.4.2 (Jacobian variety of a curve). (♠ TODO: separate out jacobian of just a proper curve and the jacobian of a nice curve) Let X be a proper (Definition 1.2.2) curve over a field (Definition A.1.3) k .

1. The relative Picard functor (Definition 3.1.1) $\mathrm{Pic}_{X/k}$ is representable by a smooth (Definition 1.2.3) group scheme (Definition 2.1.1) over k . The *(generalized) Jacobian of X* refers to the identity component (Definition 2.1.2) $\mathrm{Pic}_{X/k}^0$ of this group scheme.
2. Now assume that X is smooth, is projective, and has geometrically connected (Definition A.2.6) connected components. The Jacobian is an abelian variety (Definition 2.3.1) over k and may be called the *Jacobian variety of X* . It may be denoted by notations such as $\mathrm{Jac}(X)$, Jac_X , or J_X .

Under these assumptions, note that $\mathrm{Jac}(X)$ coincides with the Picard variety (Definition 3.3.1) of X/k . Equivalently, J_X parametrizes isomorphism classes of degree-zero line bundles (Definition A.3.5) on X , where a line bundle $\mathcal{L}$ has *degree zero* if its degree (computed as the sum of orders of any rational function defining it) vanishes.

The Jacobian variety $\mathrm{Jac}(X)$ has dimension g where g is the geometric genus of X .

When X is equipped with a base point, the Jacobian variety carries a canonical principal polarization (Definition 2.3.6) Θ , making (J_X, Θ) a principally polarized abelian variety.

Proposition 3.4.3 (see e.g. [Mil08, Chapter III, Proposition 2.1.]). (♠ TODO: tangent space) Let X be a smooth (Definition 1.2.3), projective curve whose connected components are geometrically connected (Definition A.2.6). The tangent space to the Jacobian variety (Definition 3.4.2) $\mathrm{Jac}(X)$ at the identity point is canonically isomorphic to $H^1(X, \mathcal{O}_X)$. Consequently, the dimension of $\mathrm{Jac}(X)$ equals the geometric genus of X .

3.5. Coverings of curves from the Jacobian.

Proposition 3.5.1 (see e.g. [Mil08, Chapter III, Proposition 9.1.]). Let X be a smooth (Definition 1.2.3), projective curve over a field (Definition A.1.3) k whose connected components are geometrically connected (Definition A.2.6).

(♠ TODO: etale covering)

1. Suppose that $J' \rightarrow \mathrm{Jac}(X)$ is a connected étale covering of the Jacobian variety (Definition 3.4.2) $\mathrm{Jac}(X)$. The pullback $X \times_{\mathrm{Jac}(X)} J' \rightarrow X$ is a connected étale covering of X .
2. Every connected abelian étale covering of X is obtained via connected étale coverings of $\mathrm{Jac}(X)$ as above.
(♠ TODO: do I need to assume the curve to be connected) (♠ TODO: define abelianization)
3. Let $P \in X(k)$. There is a map $\pi_1^{\mathrm{ét}}(X, P)^{\mathrm{ab}} \rightarrow \pi_1^{\mathrm{ét}}(\mathrm{Jac}(X), 0)$?? induced by this choice of P and it is an isomorphism where 0 is the identity point of $\mathrm{Jac}(X)$.

Proposition 3.5.2. (♠ TODO: talk about how

$$H^1(X_{\text{et}}, G) \cong \text{Hom}_{\text{cont}}(\pi_1^{\text{et}}(X, \bar{x}), G)$$

for finite groups G and general connected schemes X .)

Proposition 3.5.3 (see [Mil08, Chapter III, Corollary 9.6.]). Let X be a smooth (Definition 1.2.3), projective curve over a field (Definition A.1.3) k whose connected components are geometrically connected (Definition A.2.6). (♠ TODO: handle the disconnectedness of X ; apparently the abel-jacobi map would need to pick out rational points on all components.) Let $P \in X(k)$. For all primes ℓ , the map of étale cohomology groups $H^1(\text{Jac}(X), \mathbb{Z}_\ell) \rightarrow H^1(X, \mathbb{Z}_\ell)$ induced by the embedding $X \hookrightarrow \text{Jac}(X)$ is an isomorphism. (♠ TODO: apparently this is still true for $\ell = p$ by a Witt vector argument)

(♠ TODO: look into SGA 1, Expose XI for result that relates étale fundamental group of abelian variety to adic tate module)

4. ABELIAN SCHEMES

Definition 4.0.1 (Abelian scheme). Let S be a scheme. An *abelian scheme over S* is a proper (Definition 1.2.2) and smooth (Definition 1.2.3) group scheme (Definition 2.1.1)

$$\pi : A \rightarrow S$$

with geometrically connected (Definition A.2.6) fibers. Each geometric fiber (Definition A.2.4) $A_{\bar{s}}$ (over a geometric point $\bar{s} \rightarrow S$) is then an abelian variety (Definition 2.3.1).

In particular, over a field, an abelian scheme is precisely an abelian variety.

Definition 4.0.2 (Isogeny of abelian schemes). Let S be a scheme, and let A and B be abelian schemes over S (Definition 4.0.1). An *isogeny of abelian schemes over S* or *S -isogeny of abelian schemes* is a morphism of group schemes

$$\varphi : A \rightarrow B$$

(♠ TODO: define finite, faithfully flat, and surjective scheme morphisms) over S which is finite, faithfully flat, and surjective.

Equivalently, φ is a morphism of abelian schemes whose geometric fibers (Definition A.2.4) are *isogenies of abelian varieties*, i.e., surjective homomorphisms with finite kernel.

The kernel of φ is often denoted by $\ker \varphi$ or $A[\varphi]$.

5. NÉRON MODELS

Definition 5.0.1 (Néron model of an abelian variety). Let S be an integral scheme (Definition 1.2.5), let η be its generic point (Definition 1.1.4), and let $K = \mathcal{O}_{S, \eta}$ be the function field (Definition A.2.3) of S . Let A_K be an abelian variety over K (Definition 2.3.1).

A *Néron model of A_K over S* is a smooth (Definition 1.2.3) separated group scheme $\mathcal{A}$ of finite type over S whose generic fiber (Definition 1.1.3) $\mathcal{A}_\eta$ is isomorphic to A_K , satisfying

the following universal property (the *Néron mapping property*): for every smooth S -scheme U and every K -morphism $f_K : U_\eta \rightarrow A_K$, there exists a unique S -morphism $f : U \rightarrow \mathcal{A}$ extending f_K .

If such a model exists, it is unique up to unique isomorphism.

6. REDUCTION OF ABELIAN VARIETIES

6.1. Good and semi-abelian reduction.

Definition 6.1.1 (Good reduction of an abelian variety over a Dedekind scheme). Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let A_K be an abelian variety over K (Definition 2.3.1), and let $s \in S$ be a closed point.

We say that A_K has *good reduction at s* if there exists a smooth and proper scheme $\mathcal{A}$ over S whose generic fiber $\mathcal{A}_\eta$ is isomorphic to A_K . In this case, the special fiber $\mathcal{A}_s = \mathcal{A} \times_S \text{Spec } k(s)$ (where $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$ is the residue field at s) is an abelian variety over $k(s)$, and we say that $\mathcal{A}_s$ is the *good reduction of A_K at s* .

Equivalently, A_K has good reduction at s (Definition 6.1.1) if and only if there is a Néron model (Definition 5.0.1) $\mathcal{A}$ of A over the local ring $\mathcal{O}_{S,s}$ (or its completion $\widehat{\mathcal{O}}_{S,s}$ is proper (Definition 1.2.2) (or equivalently an abelian scheme (Definition 4.0.1)).

Definition 6.1.2 (Abelian and semi-abelian reduction for a smooth group scheme over a Dedekind scheme). Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Assume for simplicity that S is connected.

Let G be a smooth (Definition 1.2.3) group scheme of finite type (Definition 2.1.1) over S . For any closed point $s \in S$, the special fiber of G at s is the group scheme $G_s = G \times_S \text{Spec } k(s)$ over the residue field $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$. Let $(G_s)^0$ denote the identity component (Definition 2.1.2) of G_s .

We say that G has *abelian reduction at s* if $(G_s)^0$ is an abelian variety over $k(s)$ (Definition 2.3.1). We say that G has *semi-abelian reduction at s* if $(G_s)^0$ fits into a short exact sequence of algebraic groups over $k(s)$:

$$1 \rightarrow T \rightarrow (G_s)^0 \rightarrow B \rightarrow 0,$$

where T is an algebraic torus over $k(s)$ (Definition 2.2.1) and B is an abelian variety over $k(s)$.

In particular, if G is a Néron model (Definition 5.0.1) of its generic fiber G_K , we say that G_K has *abelian (resp. semi-abelian) reduction at s* if the corresponding fact holds for G . This is equivalent to saying that the base change $G \times_S \text{Spec } \mathcal{O}_{S,s}$ of G to $\text{Spec } \mathcal{O}_{S,s}$ has abelian (resp. semi-abelian) reduction at the unique closed point of $\text{Spec } \mathcal{O}_{S,s}$.

6.2. Potential reduction.

Definition 6.2.1 (Potential good and semi-abelian reduction for an abelian variety over the function field of a Dedekind scheme). Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point, and let $K = \mathcal{O}_{S,\eta}$ be the function field of S . Let A_K be an abelian variety over K (Definition 2.3.1), and let $s \in S$ be a closed point.

We say that A_K has *potential good reduction at s* (resp. *potential abelian reduction at s* , resp. *potential semi-abelian reduction at s*) if there exists a finite Galois extension (Definition A.1.4) L of K such that $A_L = A_K \times_K L$ has abelian (resp. semi-abelian) reduction at all closed points $s' \in S'$ lying over s , where S' is the normalization of S in L . Instead of potential abelian reduction, we will also talk about *potential good reduction*.

7. GALOIS REPRESENTATIONS AND TATE MODULES

7.1. Absolute Galois groups and decomposition theory.

Definition 7.1.1 (Separable closure of a field). Let K be a field (Definition A.1.3). A *separable closure of K* is a separable algebraic extension K^{sep} of K such that every element of K^{sep} is separable over K , and K^{sep} has no nontrivial separable algebraic extensions (i.e., the only separable algebraic extension of K^{sep} inside a fixed algebraic closure $\bar{K}$ is itself).

K^{sep} is often also denoted as K^s , K_s , or $\bar{K}$, the latter of which should not be mistaken for notation for an algebraic closure of K .

(♠ **TODO: compositum of fields**) Equivalently, K^{sep} is the compositum of all finite separable subextensions of K in a fixed algebraic closure $\bar{K}$. A separable closure always exists and is unique up to (non-unique) isomorphism over K . If K has characteristic (Definition A.1.2) zero, then every algebraic extension is separable, so the separable closure coincides with the algebraic closure.

Definition 7.1.2 (Absolute Galois group of a field). Let K be a field (Definition A.1.3), and let K^s be a separable closure (Definition 7.1.1) of K . Write $\bar{K}$ for an algebraic closure of K containing K^s .

The *absolute Galois group of K* is the Galois group (Definition A.1.4):

$$G_K = \text{Gal}(K^s/K)$$

equipped with the Krull topology (Definition A.1.5), making it a profinite group. Equivalently, G_K is the inverse limit (Definition A.1.6) of the finite Galois groups $\text{Gal}(L/K)$ as L ranges over all finite Galois subextensions of K^s/K :

$$G_K = \varprojlim_L \text{Gal}(L/K)$$

We note that $\overline{\text{Aut}}(\bar{K}/K)$ is naturally isomorphic as a profinite group to the absolute Galois group $\text{Gal}(K^s/K)$. One often writes $\text{Gal}(\bar{K}/K)$ for $\overline{\text{Aut}}(\bar{K}/K)$, even when $\bar{K}/K$ is not a Galois extension (which is the case when K is not a perfect field). On the other hand, if K is a perfect field, then the separable closure K^s coincides with the algebraic closure $\bar{K}$, and $\bar{K}/K$ is a Galois extension, so the notation of $\text{Gal}(\bar{K}/K)$ is no longer an abuse of notation.

Definition 7.1.3 (Decomposition group of a Dedekind scheme at a closed point). Let S be a Dedekind scheme (Definition 1.1.1) with function field K . Fix a separable closure (Definition 7.1.1) $\bar{K}$ of K , and let $G_K = \text{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition 7.1.2) of K . Let $s \in S$ be a closed point (Definition 1.1.5).

Choosing a prime $\bar{\mathfrak{s}}$ of the integral closure (Definition A.2.2) of $\mathcal{O}_{S,s}$ in $\bar{K}$ is equivalent to choosing a geometric point $\bar{s} \rightarrow S$ lying over s . The *decomposition group of S at s (with respect to this choice)* is the subgroup:

$$D_s = D(\bar{\mathfrak{s}}|s) = \{\sigma \in G_K \mid \sigma(\bar{\mathfrak{s}}) = \bar{\mathfrak{s}}\} \subset G_K$$

The decomposition group D_s is well-defined up to conjugacy in G_K . There is a natural surjection $\phi : D_s \twoheadrightarrow \text{Gal}(\overline{k(s)}/k(s))$, where $\overline{k(s)}$ is a separable closure of the residue field $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$. The kernel of this surjection is the inertia subgroup.

Definition 7.1.4 (Inertia group in the absolute Galois group of the function field of a Dedekind scheme at a closed point). Let S be a Dedekind scheme (Definition 1.1.1) with function field K . Fix a separable closure (Definition 7.1.1) $\bar{K}$ of K , and let $G_K = \text{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition 7.1.2) of K . Let $s \in S$ be a closed point (Definition 1.1.5) with residue field (Definition 1.2.9) $k(s) = \mathcal{O}_{S,s}/\mathfrak{m}_s$.

Choosing a prime $\bar{\mathfrak{s}}$ of the integral closure of $\mathcal{O}_{S,s}$ in $\bar{K}$ is equivalent to choosing a geometric point $\bar{s} \rightarrow S$ lying over s . This choice determines the decomposition group (Definition 7.1.3) $D_s = D(\bar{\mathfrak{s}}|s) \subset G_K$, together with a natural surjection $\phi : D_s \twoheadrightarrow \text{Gal}(\overline{k(s)}/k(s))$, where $\overline{k(s)}$ is a separable closure of $k(s)$.

The *inertia group in the absolute Galois group of the function field of S at s (with respect to the choice of $\bar{\mathfrak{s}}$)* is the subgroup:

$$I_s = I(\bar{\mathfrak{s}}|s) = \ker \left(D_s \xrightarrow{\phi} \text{Gal}(\overline{k(s)}/k(s)) \right) \subset G_K$$

Equivalently, I_s consists of all automorphisms $\sigma \in G_K$ that induce the identity on the residue field (Definition 1.2.9) of the valuation ring (Definition A.2.1) corresponding to $\bar{\mathfrak{s}}$.

7.2. Tate modules and unramified representations.

Definition 7.2.1 (Tate module of an abelian variety). (♠ TODO: separable closure) Let K be a field (Definition A.1.3), let $\bar{K}$ be a separable closure of K , and let $G_K = \text{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition A.1.4) of K . Let A_K be an abelian variety (Definition 2.3.1) over K , and let ℓ be a prime number (Definition A.1.1) that is distinct from the characteristic (Definition A.1.2) of K .

For any integer $\nu \geq 1$, let $A_K[\ell^\nu]$ be ℓ^ν -torsion subgroup of A_K , which is a finite étale group scheme over K (or equivalently, as a discrete G_K -module via its $\bar{K}$ -rational points). The *ℓ -adic Tate module of A_K* is the G_K -module:

$$T_\ell(A_K) = \varprojlim_{\nu} A_K[\ell^\nu](\bar{K})$$

where the inverse limit is taken with respect to the multiplication-by- ℓ transition maps. One may also consider the rational Tate module:

$$V_\ell(A_K) = T_\ell(A_K) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

which is a vector space over $\mathbb{Q}_\ell$ of dimension equal to the relative dimension of A_K , equipped with a continuous linear action of G_K .

Definition 7.2.2 (Unramified Galois representation at a closed point). Let S be a Dedekind scheme (Definition 1.1.1) with function field (Definition A.2.3) K . Fix a separable closure (Definition 7.1.1) $\bar{K}$ of K , and let $G_K = \text{Gal}(\bar{K}/K)$ be the absolute Galois group (Definition 7.1.2) of K . Let $s \in S$ be a closed point (Definition 1.1.5), and fix a prime $\bar{s}$ of the integral closure (Definition A.2.2) of $\mathcal{O}_{S,s}$ in $\bar{K}$. Let $I_s \subset G_K$ be the corresponding inertia subgroup (Definition 7.1.4).

Let M be a continuous (Definition A.4.2) G_K -module (such as a finite torsion module or an ℓ -adic Tate module (Definition 7.2.1)). We say that M is *unramified at s* if the inertia group (Definition 7.1.4) I_s acts trivially (Definition A.4.1) on M , i.e.,

$$\sigma \cdot m = m \quad \text{for all } \sigma \in I_s \text{ and all } m \in M.$$

In particular, for an abelian variety A_K over K and a prime ℓ distinct from the characteristic of $k(s)$, we say that the Tate module $T_\ell(A_K)$ (resp. $V_\ell(A_K)$) is *unramified at s* if I_s acts trivially on $T_\ell(A_K)(\bar{K})$ (resp. $V_\ell(A_K)$).

Remark 7.2.3. Although the inertia group I_s depends on the choice of $\bar{s}$, the property of being unramified at s does not: all inertia groups at s are conjugate in G_K , and since any continuous G_K -module is stable under the full Galois group, if one inertia subgroup acts trivially on M , then all do.

8. THE CRITERION OF NÉRON–OGG–SHAFAREVICH

Theorem 8.0.1 (Criterion of Néron–Ogg–Shafarevich for good reduction of abelian varieties). Let S be a Dedekind scheme (Definition 1.1.1), let η be its generic point (Definition 1.1.4), and let $K = \mathcal{O}_{S,\eta}$ be the function field (Definition A.2.3) of S . Let A_K be an abelian variety over K (Definition 2.3.1), and let $s \in S$ be a closed point (Definition 1.1.5). Fix a prime $\bar{s}$ of the integral closure (Definition A.2.2) of $\mathcal{O}_{S,s}$ in a separable closure (Definition 7.1.1) $\bar{K}$ of K , and let $I_s \subset G_K = \text{Gal}(\bar{K}/K)$ be the corresponding inertia subgroup (Definition 7.1.4). Let ℓ be a prime number (Definition A.1.1) distinct from the characteristic (Definition A.1.2) of the residue field (Definition 1.2.9) $k(s)$.

The following conditions are equivalent:

1. A_K has good reduction at s (Definition 6.1.1).
2. There exists an abelian scheme (Definition 4.0.1) $\mathcal{A}$ over S whose generic fiber (Definition 1.1.3) is isomorphic to A_K . In this case, the Néron model (Definition 5.0.1) of A_K over S is an abelian scheme over S , and its special fiber $\mathcal{A}_s$ is an abelian variety over $k(s)$.

3. The inertia group I_s acts trivially on the ℓ^ν -torsion points $A_K[\ell^\nu](\bar{K})$ for some integer $\nu \geq 1$.
4. The inertia group I_s acts trivially on the full ℓ -adic Tate module (Definition 7.2.1) $T_\ell(A_K)(\bar{K})$.
5. The inertia group I_s acts trivially on the rational ℓ -adic Tate module $V_\ell(A_K) = T_\ell(A_K) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.
6. The Tate module $T_\ell(A_K)$ (equivalently, $V_\ell(A_K)$) is an unramified (Definition 7.2.2) continuous G_K -representation at s .

Moreover, if A_K has good reduction (Definition 6.1.1) at s with abelian scheme model $\mathcal{A}$, then for every prime ℓ' distinct from $\text{char}(k(s))$, the Tate module $T_{\ell'}(A_K)$ is unramified at s . Furthermore, for any such ℓ' , the specialization map induces an isomorphism of $G_{k(s)}$ -modules:

$$T_{\ell'}(A_K) \xrightarrow{\sim} T_{\ell'}(\mathcal{A}_s)$$

where the $G_{k(s)} = \text{Gal}(\bar{k(s)}/k(s))$ -action on the left is via the identification $G_{k(s)} \cong D_s/I_s$.

9. SELMER THEORY

Definition 9.0.1. Assume ?? and further assume that L/K is a finite Galois extension (Definition A.1.4).

The *decomposition group of L/K* is the subgroup

$$D_w = \{\sigma \in \text{Gal}(L/K) : w \circ \sigma = w\}$$

(Definition A.1.4)

Definition 9.0.2. Assume ?? and further assume that L/K is a finite Galois extension (Definition A.1.4).

There is a natural surjective homomorphism

$$D_w \rightarrow \text{Gal}(k_w/k_v)$$

on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_w} \text{ for all } x \in \mathcal{O}_w\}$$

(Definition 9.0.1)

Definition 9.0.3. Assume ?? and further assume that L/K is a Galois extension (Definition A.1.4).

The *decomposition group of L/K* is the subgroup

$$D_w = \{\sigma \in \text{Gal}(L/K) : w \circ \sigma = w\}.$$

(Definition A.1.4) It is also the inverse limit (Definition A.1.6) of the decomposition groups (Definition 9.0.1) of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of $\text{Gal}(L/K)$.

Letting (K, v) be a valuation field, its *decomposition group* $D_{K,v}$ (also sometimes denoted by $G_{K,v}$) is the decomposition group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition 9.0.4. Assume ?? and further assume that L/K is a Galois extension (Definition A.1.4).

There is a natural surjective homomorphism

$$D_w \rightarrow \text{Gal}(k_w/k_v)$$

(Definition 9.0.3) on the residue fields. The *inertia group of L/K* is its kernel:

$$I_w = \{\sigma \in D_w : \sigma(x) \equiv x \pmod{\mathfrak{m}_v} \text{ for all } x \in \mathcal{O}_v\}.$$

(Definition 9.0.3) (Definition A.1.4) It is also the inverse limit (Definition A.1.6) of the decomposition groups (Definition 9.0.1) of the finite Galois subextensions E/K within L (with valuation restricted from the valuation of L). It is a closed subgroup of D_w .

Letting (K, v) be a valuation field, its *inertia group* $I_{K,v}$ is the inertia group of K^{sep}/K in the above sense where a valuation on K^{sep} extending v is fixed.

Definition 9.0.5 (Kummer map associated to an isogeny of an abelian variety). Let k be a field (Definition A.1.3), and let A and B be commutative group schemes (Definition 2.1.1) over S . Suppose $\varphi : A \rightarrow B$ is an isogeny over S . Denote by $G_k = \text{Gal}(\bar{k}/k)$ the absolute Galois group (Definition 7.1.2) of k .

The short exact sequence

$$0 \rightarrow \ker \varphi \rightarrow A(\bar{k}) \xrightarrow{\varphi} B(\bar{k}) \rightarrow 0.$$

yields the following long exact sequence in Galois cohomology:

$$\begin{aligned} 0 \rightarrow \ker \varphi(k) \rightarrow A(k) \xrightarrow{\varphi} B(k) \\ \xrightarrow{\delta_\varphi} H^1(G_k, \ker \varphi) \rightarrow H^1(G_k, A(\bar{k})) \rightarrow H^1(G_k, B(\bar{k})) \rightarrow \dots \end{aligned}$$

The *Kummer map associated to the isogeny φ* is the Galois cohomological connecting homomorphism

$$\delta_\varphi : B(k) \rightarrow H^1(G_k, \ker \varphi),$$

above. In fact, there is a short exact sequence

$$0 \rightarrow B(k)/\varphi(A(k)) \xrightarrow{\delta'_\varphi} H^1(G_k, \ker \varphi) \rightarrow H^1(G_k, A)[\varphi] \rightarrow 0$$

where the map δ'_φ is induced by δ_φ . We may also let the *Kummer map associated to φ* refer to this map δ'_φ . We may also use the abuse of notation δ_φ to denote δ'_φ .

Definition 9.0.6 (Selmer group for an isogeny of abelian group schemes over a global field). (♠ TODO: It should be possible to define this for more general group schemes) Let K be a global field, and let A and B be abelian group schemes (Definition 2.1.1) defined over K . Suppose $\varphi : A \rightarrow B$ is an isogeny defined over K .

Denote by $G_K = \text{Gal}(\overline{K}/K)$ the absolute Galois group (Definition 7.1.2) of K . For each place v of K , fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group (Definition 9.0.3) $G_v \subset G_K$.

The *Selmer group of φ over K* , denoted by notations such as $\text{Sel}(A/K)$, $\text{Sel}^\varphi(A/K)$, $\text{Sel}^{(\varphi)}(A/K)$, $\text{Sel}_\varphi(A/K)$, is the subgroup of the Galois cohomology group $H^1(K, \ker \varphi)$ given by

$$\text{Sel}^\varphi(A/K) := \ker \left(H^1(G_K, A[\varphi]) \rightarrow \prod_v H^1(G_v, A(\overline{K}))[\varphi] \right),$$

(♠ TODO: define the kummer map associated to φ) where the product runs over all places v of K , and $H^1(G_v, A)[\varphi]$ denotes the image of the local Kummer map associated to φ .

We describe the map

$$(A) \quad H^1(G_K, A[\varphi]) \rightarrow \prod_v H^1(G_v, A(\overline{K}))[\varphi]$$

(♠ TODO: define a decomposition group) used to define the kernel above: Note that G_v acts on $A(\overline{K}_v)$ and $B(\overline{K}_v)$, and the base change φ_v of φ to K_v induces a Kummer short exact sequence (Definition 9.0.5)

$$0 \rightarrow B(K_v)/\varphi(A(K_v)) \xrightarrow{\delta_{\varphi_v}} H^1(G_v, A[\varphi]) \rightarrow H^1(G_v, A(\overline{K}_v))[\varphi] \rightarrow 0.$$

The natural inclusions $G_v \subset G_K$ and $E(K) \subset E(K_v)$ give restriction maps $H^1(G_K, A[\varphi]) \rightarrow H^1(G_v, A[\varphi])$ on Galois cohomology, and we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & B(K)/\varphi(A(K)) & \xrightarrow{\delta_\varphi} & H^1(G_K, A[\varphi]) & \longrightarrow & H^1(G_K, A(\overline{K}))[\varphi] \longrightarrow 0 \\ & & \downarrow & & \downarrow \Pi_v \text{res}_v & \searrow \text{dotted} & \downarrow \\ 0 & \longrightarrow & \prod_v B(K_v)/\varphi(A(K_v)) & \xrightarrow{\delta_{\varphi_v}} & \prod_v H^1(G_v, A[\varphi]) & \longrightarrow & \prod_v H^1(G_v, A(\overline{K}_v))[\varphi] \longrightarrow 0 \end{array}$$

The map (A) is the one given in the above commutative diagram by the dotted arrow.

Equivalently,

$$\text{Sel}^\varphi(A/K) = \{c \in H^1(K, A[\varphi]) : \text{for all places } v \text{ of } K, \text{res}_v(c) \in \text{Im } \delta_{\varphi_v}\}$$

(Proposition 9.0.7)

Proposition 9.0.7. Let K be a global field, and let A and B be abelian group schemes (Definition 2.1.1) defined over K . Let $\varphi : A \rightarrow B$ be an isogeny defined over K . Denote by $G_K = \text{Gal}(\overline{K}/K)$ the absolute Galois group (Definition 7.1.2) of K . For each place v of K , fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group (Definition 9.0.3) $G_v \subset G_K$. Write $\text{res}_v : H^1(K, A[\varphi]) \rightarrow H^1(G_v, A[\varphi])$ for the restriction maps induced by the inclusions $G_v \subset G_K$.

The Selmer group (Definition 9.0.6) $\text{Sel}(A/K)$ coincides with the following group:

$$\left\{ c \in H^1(K, A[\varphi]) : \text{for all places } v \text{ of } K, \text{res}_v(c) \in \text{Image} \left(\frac{B(K_v)}{\varphi(A(K_v))} \xrightarrow{\delta_{\varphi_v}} H^1(G_v, A[\varphi]) \right) \right\}$$

(??) (??)

Proof. By the exactness of

$$0 \longrightarrow \prod_v B(K_v)/\varphi(A(K_v)) \xrightarrow{\delta_{\varphi_v}} \prod_v H^1(G_v, A[\varphi]) \xrightarrow{\pi} \prod_v H^1(G_v, A(\overline{K}_v))[\phi] \longrightarrow 0,$$

the image of δ_{φ_v} equals the kernel of π . By definition, the Selmer group is the kernel of the composition of π with

$$H^1(G_K, A[\varphi]) \rightarrow \prod_v H^1(G_v, A[\varphi])$$

and hence the Selmer group is contained in the kernel of π . Inversely, if the $c \in H^1(G_K, A[\varphi])$ is not an element of the Selmer group, then it is by definition sent to a nonzero element of $\prod_v H^1(G_v, A(\overline{K}_v))[\phi]$, which implies that its image under $\prod_v \text{res}_v$ cannot be an element of $\ker \pi = \text{Im } \delta_{\varphi_v}$. $\square$

Proposition 9.0.8 ([Če16, Proposition 2.7(d)]). (♠ TODO: figure out how to have local tamagawa numbers applicable to this setting) Let $S = \text{Spec } \mathfrak{o}$ (Definition 1.2.4) where $\mathfrak{o}$ is a Henselian discrete valuation ring with fraction field K and residue field k . Let $\varphi : A \rightarrow B$ be a K -isogeny (Definition 2.3.4) of abelian varieties (Definition 2.3.1). Write $\delta_{\varphi} : B(K) \rightarrow H^1(K, A[\varphi])$ (Definition 2.1.6) for the Kummer (Definition 9.0.5) map associated to φ . Let $\mathcal{A}/S$ and $\mathcal{B}/S$ be the Néron models (Definition 5.0.1) of A and B respectively, and lift $\varphi : A \rightarrow B$ to an isogeny $\varphi : \mathcal{A} \rightarrow \mathcal{B}$. Further write $\varphi : (\mathcal{A}_k/\mathcal{A}_k^0) \rightarrow (\mathcal{B}_k/\mathcal{B}_k^0)$ (Definition 2.1.2) for the induced homomorphism between the étale k -group schemes of connected components of the fibers $\mathcal{A}_k$ and $\mathcal{B}_k$. Let c_A and c_B respectively be the local Tamagawa numbers of A and B .

Assuming that

1. $\mathcal{A}[\varphi] \rightarrow S$ is étale (which holds whenever $\text{Char } k \nmid \deg \varphi$ for example (♠ TODO: ref)), and
2. We have

$$\#(\mathcal{A}_k/\mathcal{A}_k^0(k)) = \#(\varphi(\mathcal{A}_k/\mathcal{A}_k^0)(k)) = \#(\mathcal{B}_k/\mathcal{B}_k^0(k))$$

(which holds whenever $\deg \varphi$ is coprime to $c_A \cdot c_B$ (♠ TODO: ref))

we have

$$H_{\text{unr}}^1(K, A[\varphi]) = \text{Im } \delta_{\varphi} = H_{\text{fppf}}^1(S, \mathcal{A}[\varphi])$$

(??) (??) (??) (♠ TODO: flat cohomology)

Corollary 9.0.9. (♠ TODO: if selmer groups can be extended to more general bases, then this should be generalizable to more general bases whose residue fields are all of a single characteristic) Let K be a global function field over a finite field $\mathbb{F}_q$, and let A and B be abelian group schemes (Definition 2.1.1) defined over K . Let $\varphi : A \rightarrow B$ be an isogeny defined over K . Denote by $G_K = \text{Gal}(\overline{K}/K)$ the absolute Galois group (Definition 7.1.2) of K . For each place v of K , fix an extension v to $\overline{K}$, which yields an embedding $\overline{K} \subset \overline{K}_v$ and a decomposition group (Definition 9.0.3) $G_v \subset G_K$. Write $\text{res}_v : H^1(K, A[\varphi]) \rightarrow H^1(G_v, A[\varphi])$

for the restriction maps induced by the inclusions $G_v \subset G_K$. Write $\delta_{\varphi_v} : B(K_v)/\varphi(A(K_v)) \rightarrow H^1(G_v, A[\varphi])$ for the local Kummer map (Definition 9.0.5) for A/K_v , and write $c_{A,v}$ and $c_{B,v}$ for the local Tamagawa numbers of A and B over the completion K_v of K at v .

If $\deg \varphi$ is coprime to $p \cdot \prod_v c_{A,v} \cdot \prod_v c_{B,v}$, then

$$\text{Sel}^\varphi = \{c \in H^1(K, A[\varphi]) : \text{for every place } v \text{ of } K, \text{res}_v(C) \in H_{\text{unr}}^1(K_v, A[\varphi])\}$$

(??)

Proof. Let S be the smooth projective curve over $\mathbb{F}_q$ corresponding to K . It is a Dedekind scheme of finite type over $\mathbb{F}_q$, so Néron models $\mathcal{A}$ and $\mathcal{B}$ of A and B over S exist. Writing $\varphi : \mathcal{A} \rightarrow \mathcal{B}$ for the isogeny lifting φ , we note that $\mathcal{A}[\varphi]$ is étale over S as a consequence of ???. This holds by a combination of Proposition 9.0.7 and Proposition 9.0.8 — letting $\mathcal{A}$ and $\mathcal{B}$ be the Néron models of A and B over the smooth projective curve S over $\mathbb{F}_q$ corresponding to K $\square$

APPENDIX A. BACKGROUND DEFINITIONS

A.1. Basic algebra and set theory.

Definition A.1.1. A **prime number** is a positive element of $\mathbb{Z}$ that is an irreducible element of $\mathbb{Z}$.

Explicitly, a prime number p is a positive integer that is not 1 and whose only positive integer divisors are 1 and p

Equivalently, a prime number is a positive element of $\mathbb{Z}$ that is a prime element of $\mathbb{Z}$.

A prime number is also called a **rational prime** to distinguish the notion of a prime number from other notions of "primes" that arise in arithmetic.

Definition A.1.2. Let R be a ring. There exists a unique ring homomorphism $\psi : \mathbb{Z} \rightarrow R$ defined by mapping $n \mapsto n \cdot 1_R$. The non-negative generator of the kernel of this map, $\ker(\psi) = \langle p \rangle \subseteq \mathbb{Z}$, is called the **characteristic of R** , denoted by $\text{char}(R)$.

Definition A.1.3 (Field). A **field** is commutative division ring. In other words, a field is a commutative ring for which all nonzero elements have a multiplicative inverse.

Definition A.1.4 (Galois Extension). An extension L/K is called a **Galois extension** if it is both a normal extension and a separable algebraic extension. Its **Galois group**, usually denoted by $\text{Gal}(L/K)$, is defined to be the automorphism group $\text{Aut}(L/K)$.

Definition A.1.5. A **profinite set** is a set which is the projective limit (Definition A.1.6) of a (small) directed system of finite sets.

A profinite set $X = \varprojlim_i X_i$ has a natural topology, referred to as the **profinite topology**, **inverse limit topology**, or **Krull topology**. It can be formulated in the following equivalent ways:

(♠ TODO: define inverse limit topology)

1. As the inverse limit (Definition A.1.6) topology with respect to the family of projection maps $\pi_i : X \rightarrow X_i$ onto the finite sets X_i , each equipped with the discrete topology, in the system.
2. As the subspace topology on $\varprojlim_i X_i$ inherited from the product topology on $\prod_i X_i$.
3. As the topology generated by the basis of "finite index" sets (specifically, the clopen sets) of the form $\pi_i^{-1}(V)$ for some index i and some subset $V \subseteq X_i$.

These topologies do not depend on the inverse system representing X .

Definition A.1.6 (Special cases of limits). Let $\mathcal{C}$ be a (large) category. Let I be a (large) category. Let $I \rightarrow \mathcal{C}$ be a diagram/system.

- Suppose that the system is a cofiltered system, i.e. I is a cofiltered category. A limit of this diagram is often denoted by

$$\varprojlim_{i \in I} D(i)$$

and may be called a *cofiltered (inverse/projective) limit*. In case that the system is more specifically an inverse/projective system, i.e. I is a cofiltered poset, the preferred term for such a limit is *inverse/projective limit*.

- Suppose that the system is a filtered system, i.e. I is a filtered category. A colimit of this diagram is often denoted by

$$\varinjlim_{i \in I} D(i)$$

and may be called a *filtered colimit* or a *direct/inductive/injective limit*. In case that the system is more specifically a direct/inductive system, i.e. I is a filtered poset, the preferred term for such a limit is *direct/inductive limit*.

Definition A.1.7 (Degree of an Extension, Finite Extension). (♠ TODO: dimension) Let L/K be a field extension. The *degree of the extension* is $[L : K] := \dim_K(L)$, i.e. the dimension of L as a K -vector space.

A.2. Schemes and topology.

Definition A.2.1 (Valuation Ring of a Valued Field). Let K be a field (Definition A.1.3) equipped with a valuation (♠ TODO: totally ordered abelian group) $v : K \rightarrow \Gamma$, where Γ is a totally ordered abelian group.

The *valuation ring* or *ring of integers of the valued field* (K, v) is the subring

$$\mathcal{O}_v := \{x \in K \mid v(x) \geq 0\},$$

where 0 is the neutral element of Γ . This ring $\mathcal{O}_v$ is a local ring with maximal ideal

$$\mathfrak{m}_v := \{x \in K \mid v(x) > 0\}.$$

Its *residue field* is defined as $\kappa_v := \mathcal{O}_v / \mathfrak{m}_v$.

Definition A.2.2 (Integral element over a ring). Let R be a commutative ring with unity.

1. Let A be an R -algebra. An element $a \in A$ is called **integral over R** if there exists a monic polynomial

$$p(x) = x^n + r_{n-1}x^{n-1} + \cdots + r_1x + r_0$$

with coefficients $r_i \in R$ such that

$$p(a) = a^n + r_{n-1}a^{n-1} + \cdots + r_1a + r_0 = 0 \quad \text{in } A.$$

2. Let A be an extension ring of R . The ring extension A/R is called an **integral extension** if every element of A is integral over R .
3. Let A be an extension ring of R . The **integral closure of R in A** , sometimes denoted by $\tilde{A}$, is the subring

$$\tilde{A} = \{a \in A : a \text{ is integral over } R\}.$$

We say that R is integrally closed in A if $\tilde{A}$ coincides with A (considered as a subring of R).

4. Let R be an integral domain with field of fractions $K = \text{Frac}(R)$. We say that R is **integrally closed** if it is integrally closed as a subring of K .

Definition A.2.3 (Function field). Let k be a field (Definition A.1.3). A field K is called a **function field over k** (or a **function field in n variables over k**) if K is a finitely generated field extension of k with transcendence degree n over k , where $n \geq 1$. Equivalently, the function field in n variables over k is isomorphic to the fraction field of the polynomial ring $k[x_1, \dots, x_n]$ where $x_1, \dots, x_n$ are indeterminate variables. Accordingly, the function field in n variables over k is often denoted as $k(x_1, \dots, x_n)$.

When $n = 1$, we say that K is a **function field of one variable over k** or an **algebraic function field over k** and denote it by $k(x)$.

Definition A.2.4. Let $f : X \rightarrow Y$ be a morphism of schemes (Definition 1.2.1). For any geometric point (either algebraic or separable) $\bar{y} : \text{Spec}(\Omega) \rightarrow Y$, the **geometric fiber of f at $\bar{y}$** , denoted $X_{\bar{y}}$, is the base change (Definition A.2.5):

$$X_{\bar{y}} := X \times_Y \text{Spec}(\Omega)$$

Note that this depends on the choice of the geometric point $\bar{y}$, not just on the image point $y \in Y$.

Definition A.2.5. Let $\mathcal{C}$ be a category. Let $f : X \rightarrow Y$ be a morphism in $\mathcal{C}$, and let $g : Y' \rightarrow Y$ be any morphism in $\mathcal{C}$ such that the fiber product $X \times_Y Y'$ exists. The **base change of f by g** is the morphism $f' : X \times_Y Y' \rightarrow Y'$ obtained by the pullback of f along g . In particular,

$$\begin{array}{ccc} X \times_Y Y' & \xrightarrow{p_1} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

is a Cartesian diagram (or pullback square) in $\mathcal{C}$.

Definition A.2.6. If X is a scheme over a field k , we say X is **geometrically connected over k** if the structure morphism $X \rightarrow \text{Spec}(k)$ has geometrically connected fibers; equivalently, if $X \times_k \text{Spec}(\bar{k})$ is connected.

A.3. Sheaves.

Definition A.3.1 (Sheaf on a site). Let $(\mathcal{C}, J)$ be a site. Let $\mathcal{A}$ be a (large) category.

1. A presheaf $\mathcal{F} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ is called a *sheaf on the site $(\mathcal{C}, J)$ valued in $\mathcal{A}$* if, for every object U of $\mathcal{C}$ and every covering sieve $S \in J(U)$, the limit

$$\varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \mathcal{F}|_{\mathcal{D}_S}(V),$$

exists and the canonical natural morphism

$$\mathcal{F}(U) \rightarrow \varprojlim_{(V \rightarrow U) \in (\mathcal{D}_S)^{\text{op}}} \mathcal{F}|_{\mathcal{D}_S}(V)$$

is an isomorphism. Here, $\mathcal{D}_S \hookrightarrow \mathcal{C}/U$ is the full downward-closed subcategory such that $\text{Ob}(\mathcal{D}_S) = \{(f : V \rightarrow U) : f \in S(V)\}$,

In particular, when we are working with a Grothendieck pretopology K on a category $\mathcal{C}$, we may speak of sheaves on the site whose Grothendieck topology is the one generated by K .

2. Given sheaves $\mathcal{F}, \mathcal{G} : \mathcal{C}^{\text{op}} \rightarrow \mathcal{A}$ on the site $(\mathcal{C}, J)$, a *morphism between the sheaves* is a morphism between $\mathcal{F}$ and $\mathcal{G}$ as presheaves.
3. Let U be a universe. A *U -sheaf* typically refers to a U -presheaf that is a sheaf for a U -site. In other words, a U -sheaf is a sheaf on a site whose underlying category is U -locally small and which has a U -small topologically generating family such that the sheaf is valued in U -sets.
4. The *sheaf category/category of $\mathcal{A}$ -valued sheaves on $\mathcal{C}$* is the (large) category defined as the full subcategory of $\text{PreShv}(\mathcal{C}, \mathcal{A})$ whose objects are the sheaves on $\mathcal{C}$ with values in $\mathcal{A}$. Common notations for the sheaf category include $\text{Shv}(\mathcal{C}, \mathcal{A})$, $\text{Shv}(\mathcal{C}, J, \mathcal{A})$, $\text{Sh}(\mathcal{C}, \mathcal{A})$, $\text{Sh}(\mathcal{C}, J, \mathcal{A})$. If the value category $\mathcal{A}$ is clear from context, then notations such as $\text{Shv}(\mathcal{C})$, $\text{Shv}(\mathcal{C}, J)$, $\text{Sh}(\mathcal{C})$, $\text{Sh}(\mathcal{C}, J)$ are also common.

Definition A.3.2 (Sheaf on a topological space). Let X be a topological space, let $\mathcal{D}$ be a category; a posteriori, it is necessary for $\mathcal{D}$ to have a terminal object for the following definition to be non-void. Let $\mathcal{F}$ be a presheaf valued in $\mathcal{D}$ on X . Then $\mathcal{F}$ is a *sheaf* if it satisfies the following additional condition (known as the *sheaf axioms*):

For every open set $U \subseteq X$ and every open cover $\{U_i\}_{i \in I}$ of U , let $\mathcal{J}$ be the diagram in the category of opens of U consisting of the inclusions $U_i \cap U_j \hookrightarrow U_i$ for all $i, j \in I$. Then $\mathcal{F}$ is a sheaf if the limit of the diagram $\mathcal{F} \circ \mathcal{J}$ exists in $\mathcal{D}$ and the natural morphism

$$\mathcal{F}(U) \rightarrow \lim_{j \in \mathcal{J}} \mathcal{F}(j)$$

is an isomorphism. More precisely, $\mathcal{J} : J \rightarrow \text{Open}(U)$ should be the functor whose index category J consists of

1. An object i for every $i \in I$ and an object (i, j) for every pair $i, j \in I$,
2. Morphisms $p_1 : (i, j) \rightarrow i$ and $p_2 : (i, j) \rightarrow j$ for every pair $i, j \in I$

and which sends the objects and morphisms as follows:

1. $\mathcal{J}(i) = U_i$
2. $\mathcal{J}(i, j) = U_i \cap U_j$
3. $\mathcal{J}(p_1) : U_i \cap U_j \hookrightarrow U_i$
4. $\mathcal{J}(p_2) : U_i \cap U_j \hookrightarrow U_j$.

In particular, taking $U = \emptyset$ and taking the empty open cover of the empty set, $\mathcal{F}(\emptyset)$ must be the terminal object of $\mathcal{D}$

In the case that $\mathcal{D}$ admits all small limits, the sheaf condition is equivalent to the following: For every open set $U \subset X$ and every open cover $\{U_i\}_{i \in I}$ of U , the following equalizer diagram is exact:

$$\mathcal{F}(U) \rightarrow \prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j).$$

Here, the morphism $\mathcal{F}(U) \rightarrow \prod_{i \in I} \mathcal{F}(U_i)$ and the two morphisms $\prod_{i \in I} \mathcal{F}(U_i) \rightrightarrows \prod_{i, j \in I} \mathcal{F}(U_i \cap U_j)$ are induced by the restriction maps $\mathcal{F}(U) \rightarrow \mathcal{F}(U_i)$ and $\mathcal{F}(U_i) \rightarrow \mathcal{F}(U_i \cap U_j)$.

In the case that $\mathcal{D}$ is some subcategory of the category of sets, the sheaf condition is equivalent to the following: For every open set $U \subseteq X$ and every open cover $\{U_i\}_{i \in I}$ of U ,

- (Locality) If $s, t \in \mathcal{F}(U)$ are such that $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$.
- (Gluing) If for each i there is $s_i \in \mathcal{F}(U_i)$ such that for all i, j one has $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, then there exists a unique $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i .

Equivalently, a sheaf on a topological space X may be defined as a sheaf on (Definition A.3.1) the site of opens on X .

Definition A.3.3. 1. Let $\mathcal{C}$ be a site, and let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition A.3.1) of (not necessarily commutative) rings on $\mathcal{C}$.

- (a) An $(\mathcal{A}, \mathcal{B})$ -bimodule (or a bimodule over $(\mathcal{A}, \mathcal{B})$) is a sheaf (Definition A.3.1) $\mathcal{M}$ of abelian groups on $\mathcal{C}$ equipped with a left $\mathcal{A}$ -module structure given by a morphism of sheaves (Definition A.3.1) of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$ -module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every object U in $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$\lambda_U(a, \rho_U(m, b)) = \rho_U(\lambda_U(a, m), b)$$

holds in $\mathcal{M}(U)$. In standard multiplicative notation where $\lambda(a, m)$ is denoted $a \cdot m$ and $\rho(m, b)$ is denoted $m \cdot b$, this condition is the associativity axiom

$$(a \cdot m) \cdot b = a \cdot (m \cdot b).$$

In particular, for every object $U \in \mathcal{C}$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$ -bimodule.

- (b) Assume that a sheafification functor

$$\mathrm{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists. For rings A and B , a A - B -bimodule or bimodule over (A, B) on $\mathcal{C}$ is a bimodule over $(\underline{A}, \underline{B})$ where $\underline{A}$ and $\underline{B}$ are the constant sheaves on $\mathcal{C}$ with values A and B respectively.

- (c) Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$ -bimodules. A $homomorphism of (\mathcal{A}, \mathcal{B})$ -bimodules (or an $(\mathcal{A}, \mathcal{B})$ -linear morphism) is a morphism of sheaves of abelian groups $f : \mathcal{M} \rightarrow \mathcal{N}$ such that for every object U of $\mathcal{C}$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

We denote the category of $(\mathcal{A}, \mathcal{B})$ -bimodules, with morphisms being morphisms of sheaves of abelian groups compatible with both the left $\mathcal{A}$ -action and the right $\mathcal{B}$ -action, by $\mathcal{A}\text{-}\mathcal{B}\text{-Mod}$ or sometimes by ${}_{\mathcal{A}}\mathrm{Mod}_{\mathcal{B}}$ (♠ TODO: talk about how bimodules can be identifies with left/right modules)

2. Let $(\mathcal{C}, J)$ be a site. Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on $(\mathcal{C}, J)$ (Definition A.3.1), i.e. $((\mathcal{C}, J), \mathcal{O})$ is a ringed site.

- (a) An $(left/right/two-sided) \mathcal{O}$ -module consists of the following data:

- A sheaf $\mathcal{F}$ of abelian groups on $(\mathcal{C}, J)$,
- for every object $U \in \mathcal{C}$, the structure of an $(left/right/two-sided) \mathcal{O}(U)$ -module on $\mathcal{F}(U)$,

such that for every morphism $f : V \rightarrow U$ in $\mathcal{C}$, the restriction map

$$\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

is $\mathcal{O}(U)$ -linear when the $\mathcal{O}(U)$ -action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

$$\mathcal{O}(U) \rightarrow \mathcal{O}(V)$$

induced by f .

- (b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$ -modules (Definition A.3.3).

A $morphism of \mathcal{O}$ -modules $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves (Definition A.3.1) of abelian groups such that, for every object $U \in \mathcal{C}$, the component map

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is $\mathcal{O}(U)$ -linear, i.e. it satisfies

$$\varphi_U(r \cdot s) = r \cdot \varphi_U(s) \quad \text{for all } r \in \mathcal{O}(U), s \in \mathcal{F}(U).$$

The collection of all $\mathcal{O}$ -modules together with their morphisms of $\mathcal{O}$ -modules forms the $category of \mathcal{O}$ -modules, denoted $\mathbf{Mod}(\mathcal{O})$. The Hom functors $\mathrm{Hom}_{\mathbf{Mod}(\mathcal{O})}$ of this category may be denoted by $\mathrm{Hom}_{\mathcal{O}}$.

In case that a sheafification functor

$$\mathrm{PreShv}(\mathcal{C}, \mathbf{Rings}) \rightarrow \mathrm{Shv}(\mathcal{C}, \mathbf{Rings})$$

exists, a left, right, two-sided $\mathcal{O}$ -module (and morphisms thereof) is equivalent to a $(\mathcal{O}, \mathbb{Z})$ -bimodule, $(\mathbb{Z}, \mathcal{O})$ -bimodule, and $(\mathcal{O}, \mathcal{O})$ -bimodule (and morphisms thereof) respectively, where $\mathbb{Z}$ is the constant sheaf of the integer ring $\mathbb{Z}$. In particular, the

category of $(\mathbb{Z}, \mathbb{Z})$ -bimodules is equivalent to the category of sheaves of abelian groups on $\mathcal{C}$.

Definition A.3.4. Let X be a topological space. Let $\mathcal{A}$ and $\mathcal{B}$ be sheaves (Definition A.3.2) of (not necessarily commutative) rings on X .

1. An $(\mathcal{A}, \mathcal{B})$ -bimodule is a sheaf $\mathcal{M}$ of abelian groups on X equipped with a left $\mathcal{A}$ -module structure given by a morphism of sheaves of sets

$$\lambda : \mathcal{A} \times \mathcal{M} \longrightarrow \mathcal{M},$$

and a right $\mathcal{B}$ -module structure given by a morphism of sheaves of sets

$$\rho : \mathcal{M} \times \mathcal{B} \longrightarrow \mathcal{M},$$

such that the actions are compatible. Specifically, for every open subset $U \subseteq X$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the equality

$$(a \cdot m) \cdot b = a \cdot (m \cdot b)$$

holds in $\mathcal{M}(U)$.

In particular, for every open subset $U \subseteq X$, the abelian group $\mathcal{M}(U)$ has the structure of an $\mathcal{A}(U) - \mathcal{B}(U)$ -bimodule.

2. Let $\mathcal{M}$ and $\mathcal{N}$ be $(\mathcal{A}, \mathcal{B})$ -bimodules. A *homomorphism of $(\mathcal{A}, \mathcal{B})$ -bimodules* is a morphism of sheaves of abelian groups $f : \mathcal{M} \rightarrow \mathcal{N}$ such that for every open subset $U \subseteq X$, every section $m \in \mathcal{M}(U)$, every $a \in \mathcal{A}(U)$, and every $b \in \mathcal{B}(U)$, the following compatibility conditions hold:

$$f_U(a \cdot m) = a \cdot f_U(m) \quad \text{and} \quad f_U(m \cdot b) = f_U(m) \cdot b.$$

We denote the category of $(\mathcal{A}, \mathcal{B})$ -bimodules by $\mathcal{A}\text{-}\mathcal{B}\text{-Mod}$.

3. Let $\mathcal{O}$ be a sheaf of (not necessarily commutative) rings on X , i.e., $(X, \mathcal{O})$ is a ringed space.

(a) An *(left/right/two-sided) $\mathcal{O}$ -module* consists of the following data:

- A sheaf $\mathcal{F}$ of abelian groups on X ,
- for every open subset $U \subseteq X$, the structure of an (left/right/two-sided) $\mathcal{O}(U)$ -module on $\mathcal{F}(U)$,

such that for every inclusion of open subsets $V \subseteq U$, the restriction map

$$\rho_{U,V} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

is $\mathcal{O}(U)$ -linear when the $\mathcal{O}(U)$ -action on $\mathcal{F}(V)$ is defined via the natural ring homomorphism

$$\mathcal{O}(U) \rightarrow \mathcal{O}(V)$$

induced by the inclusion.

- (b) Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}$ -modules. A *morphism of $\mathcal{O}$ -modules* is a morphism of sheaves of abelian groups $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ such that, for every open subset $U \subseteq X$, the component map

$$\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

is $\mathcal{O}(U)$ -linear.

The collection of all $\mathcal{O}$ -modules together with their morphisms forms the *category of $\mathcal{O}$ -modules*, denoted by $\mathbf{Mod}(\mathcal{O})$.

These notions are special cases of those of Definition A.3.3, where the site is taken to be the category of open subsets of X with the usual Grothendieck topology.

Definition A.3.5. Let $((\mathcal{C}, J), \mathcal{O})$ be a ringed site, where $\mathcal{O}$ is a sheaf (Definition A.3.1) of rings on the site $(\mathcal{C}, J)$. (♠ TODO: This needs some genuine fixing to be definable on a general site, not just one given by a pretopology)

1. Let I be an indexing set. The *free sheaf of $\mathcal{O}$ -modules of rank I* (or simply a *free sheaf*), denoted by $\mathcal{O}^{\oplus I}$, is the sheaf associated to the presheaf $U \mapsto \mathcal{O}(U)^{\oplus I}$. If I is finite with cardinality n , we usually write $\mathcal{O}^{\oplus n}$.
2. Let $\mathcal{A}$ be a (large) category, and let A be an object of $\mathcal{A}$. Assume that a sheafification functor (♠ TODO: If such a sheafification functor exist, does a sheafification functor exist when restricted to an object U ?)

$$a : \text{PreShv}(\mathcal{C}, J, \mathcal{A}) \rightarrow \text{Shv}(\mathcal{C}, J, \mathcal{A})$$

(Definition A.3.1) exists. Let U be an object of $\mathcal{C}$.

An $\mathcal{O}$ -module (Definition A.3.3) $\mathcal{F}$ is called *locally free of rank n on U* (for an integer $n \geq 0$) if there exists a covering sieve $\{U_i \rightarrow U\}_{i \in I}$ such that for each i , the restriction $\mathcal{F}|_{U_i}$ is isomorphic to the free sheaf $(\mathcal{O}|_{U_i})^{\oplus n}$ as an $\mathcal{O}|_{U_i}$ -module.

We might call a locally free $\mathcal{O}$ -module of rank n an *algebraic vector bundle of rank n* . A *(algebraic) line bundle* or *invertible sheaf* or is then an algebraic vector bundle of rank 1.

Definition A.3.6 (Tensor product of sheaves of bimodules). Let $\mathcal{C}$ be a site. Let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be (not necessarily commutative) sheaves (Definition A.3.1) of rings on $\mathcal{C}$. Let $\mathcal{M}$ be an $(\mathcal{R}, \mathcal{S})$ -bimodule (Definition A.3.3), and let $\mathcal{N}$ be an $(\mathcal{S}, \mathcal{T})$ -bimodule.

(♠ TODO: Consolidate with)

1. Define the *presheaf tensor product* $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$ to be the presheaf of abelian groups given by

$$U \mapsto \mathcal{M}(U) \otimes_{\mathcal{S}(U)} \mathcal{N}(U).$$

As a presheaf, it has the structure of a $\mathcal{R} - \mathcal{T}$ -bimodule (♠ TODO: bimodule for a presheaf).

2. Assume that a sheafification functor

$$\text{Shv}(\mathcal{C}, \mathbf{Sets}) \rightarrow \text{PreShv}(\mathcal{C}, \mathbf{Sets})$$

exists. The *tensor product of sheaves $\mathcal{M}$ and $\mathcal{N}$* , denoted $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$, is defined to be the sheafification of the presheaf $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$. This tensor product becomes naturally an $(\mathcal{R}, \mathcal{T})$ -bimodule.

The left $\mathcal{R}$ -action and right $\mathcal{T}$ -action are induced by the presheaf actions defined section-wise by

$$\begin{aligned} r \cdot (m \otimes n) &= (r \cdot m) \otimes n, \\ (m \otimes n) \cdot t &= m \otimes (n \cdot t), \end{aligned}$$

for all $r \in \mathcal{R}(U)$, $t \in \mathcal{T}(U)$, $m \in \mathcal{M}(U)$, and $n \in \mathcal{N}(U)$. The sheafification functor preserves these actions, yielding the required bimodule structure on $\mathcal{M} \otimes_{\mathcal{S}} \mathcal{N}$.

Inductively, given sheaves of rings $\mathcal{R}_0, \dots, \mathcal{R}_k$ and $(\mathcal{R}_{i-1}, \mathcal{R}_i)$ -bimodules $\mathcal{M}_i$ for $i = 1, \dots, k$, we may speak of the tensor product

$$\mathcal{M}_0 \otimes_{\mathcal{R}_1} \mathcal{M}_1 \otimes_{\mathcal{R}_2} \dots \otimes_{\mathcal{R}_{k-1}} \mathcal{M}_k.$$

Tensor products of sheaves are associative up to canonical isomorphism(♠ TODO:), allowing us to omit parentheses. This iterated tensor product carries a natural $(\mathcal{R}_0, \mathcal{R}_k)$ -bimodule structure.

Given a sheaf of rings $\mathcal{R}$ and a two-sided $\mathcal{R}$ -module $\mathcal{M}$, we may also speak of the *n-fold tensor product* $\mathcal{M}^{\otimes n} = \mathcal{M}^{\otimes_{\mathcal{R}} n}$.

Definition A.3.7 (Sheafified Hom). Let $(\mathcal{D}, K)$ be a site that admits a small topologically generating family (e.g. by being essentially small) and let $\mathcal{R}, \mathcal{S}, \mathcal{T}$ be sheaves of rings on $(\mathcal{D}, K)$. Let $\mathcal{M}$ be a sheaf (Definition A.3.1) of $\mathcal{R}$ - $\mathcal{S}$ bimodules (Definition A.3.3) and $\mathcal{N}$ be a sheaf of $\mathcal{R}$ - $\mathcal{T}$ bimodules.

The *sheafified Hom*, denoted by notations such as $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, $\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, or $\underline{\text{Hom}}_{\mathcal{R}}(\mathcal{M}, \mathcal{N})$, is the sheaf of $\mathcal{S}$ - $\mathcal{T}$ bimodules defined section-wise by:

$$\mathcal{H}om_{\mathcal{R}}(\mathcal{M}, \mathcal{N})(U) = \text{Hom}_{\text{Sh}(\mathcal{D}/U, K|_U; \text{Mod}(\mathcal{R}|_U))}(\mathcal{M}|_U, \mathcal{N}|_U).$$

(??) (??) (??)

Equivalently, it is the unique sheaf which satisfies the right adjoint property relative to the sheafified tensor product (??).

Definition A.3.8. Let $(\mathcal{C}, J, \mathcal{O})$ be a ringed site. The *multiplicative group sheaf of the ringed site* is the sheaf (Definition A.3.1) $\mathcal{O}^{\times}$ of groups on $(\mathcal{C}, J)$ given by

$$\mathcal{O}^{\times}(U) = \mathcal{O}(U)^{\times}$$

whose group structure is the multiplication inherited from the ring $\mathcal{O}(U)$.

The sheaf is also called the *unit sheaf of the ringed site* and is also denoted by $\mathbb{G}_m$.

A.4. Galois modules and representations.

Definition A.4.1. Let G be a group and R a not necessarily commutative ring.

1. A *trivial G -module over R* or a *trivial $R[G]$ -module* is an $R[G]$ -module (say left) A on which G acts trivial, i.e. $ga = a$ for all $g \in G$ and $a \in A$.
2. Given a (left/right) R -module A , its *trivial G -module (over R)* is the $R[G]$ -module structure whose underlying R -module is A and whose G -action is trivial. This construction yields additive functors

$$\begin{aligned} {}_R \text{Mod} &\rightarrow {}_{R[G]} \text{Mod} \\ \text{Mod}_R &\rightarrow \text{Mod}_{R[G]} \end{aligned}$$

Definition A.4.2. Let R and S be topological rings. A *topological (R, S) -bimodule* is an (R, S) -bimodule M equipped with a topology such that:

1. M is a topological abelian group under addition.

2. The left action map $R \times M \rightarrow M$, defined by $(r, m) \mapsto rm$, is continuous.
3. The right action map $M \times S \rightarrow M$, defined by $(m, s) \mapsto ms$, is continuous.

Here, the product sets $R \times M$ and $M \times S$ are endowed with the product topology.

One also calls a topological module a *continuous module*.

A *left topological R -module* is a topological $(R, \mathbb{Z})$ -bimodule, where $\mathbb{Z}$ carries the discrete topology. Similarly, a *right topological R -module* is a topological $(\mathbb{Z}, R)$ -bimodule, where $\mathbb{Z}$ carries the discrete topology. Furthermore, a *two-sided topological R -module* is a topological (R, R) -bimodule.

The discrete topology on $\mathbb{Z}$ ensures that the required continuity of the $\mathbb{Z}$ -action is satisfied by any topological abelian group M , as the map $\mathbb{Z} \times M \rightarrow M$ is continuous if and only if for each $n \in \mathbb{Z}$, the map $m \mapsto nm$ is continuous.

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